INCH-POUND
MIL-M-38510/152C
23 FEBRUARY 2005
SUPERSEDING
MIL-M-38510/152B
21 JULY 1986

MILITARY SPECIFICATION

# MICROCIRCUITS, DIGITAL, TTL, DATA DECODERS/DEMULTIPLEXERS, MONOLITHIC SILICON

Inactive for new design after 21 July 1986

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and MIL-PRF-38535

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, TTL, data decoders/demultiplexers, logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number. For this product, the requirements of MIL-M-38510 have been superseded by MIL-PRF-38535, (see 6.3)

1.2 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-38535, and as specified herein.

1.2.1 Device types. The device types should be as follows:

| Device type | Circuit |
|---|---|
| 01 | 4-line to 16-line decoder/demultiplexer |
| 02 | Dual 2-line to 4-line decoder/demultiplexer |
| 03 | Dual 2-line to 4-line decoder/demultiplexer, open collector |
| 04 | Binary-to-octal decoder |
| 05 | BCD-to-decimal decoder |
| 06 | BCD-to-decimal decoder |

1.2.2 Device class. The device class is the product assurance level as defined in MIL-PRF-38535.

1.2.3 Case outlines. The case outlines are as designated in MIL-STD-1835 and as follows:

| Outline letter | Descriptive designator | Terminals | Package style |
|---|---|---|---|
| A 1/ | GDFP5-F14 or CDFP6-F14 | 14 | Flat pack |
| B 1/ | GDFP4-F14 | 14 | Flat pack |
| C | GDIP1-T14 or CDIP2-T14 | 14 | Dual-in-line |
| D | GDFP1-F14 or CDFP2-F14 | 14 | Flat pack |
| E | GDIP1-T16 or CDIP2-T16 | 16 | Dual-in-line |
| F | GDFP2-F16 or CDFP3-F16 | 16 | Flat pack |
| J | CDIP2-T24 | 24 | Dual-in-line |
| K | GDFP2-F24 or CDFP3-F24 | 24 | Flat pack |
| Z 1/ | GDFP7-F24 or CDFP8-F24 | 24 | Flat pack |

1/ Inactive package case outline.

Comments, suggestions, or questions on this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAS, 3990 East Broad St., Columbus, OH 43218-3990, or email mailto:bipolar@dscc.dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil .

AMSC N/A FSC 5962

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1.3 Absolute maximum ratings.

| | |
|---|---|
| Supply voltage range | -0.5 V dc to 7.0 V dc |
| Input voltage range | -1.5 V dc at -12 mA to 5.5 V dc |
| Storage temperature range | -65°C to 150°C |
| Maximum power dissipation per gate ($P_D$) 1/ : | |
| Device type 01 | 270 mW |
| Device type 02 | 195 mW |
| Device type 03 | 195 mW |
| Device type 04 | 140 mW |
| Device type 05 | 190 mW |
| Device type 06 | 190 mW |
| Lead temperature (soldering, 10 seconds) | +300°C |
| Thermal resistance, junction-to-case ($\theta_{JC}$): | |
| Cases A, B, C, D, E, F, J, K, and Z | See MIL-STD-1835 |
| Junction temperature ($T_J$) | +175°C 2/ |

1.4 Recommended operating conditions.

| | |
|---|---|
| Supply voltage ($V_{CC}$) | 4.5 V dc minimum to 5.5 V dc maximum |
| Minimum high level input voltage ($V_{IH}$) | 2.0 V dc |
| Maximum low level input voltage ($V_{IL}$) | 0.8 V dc |
| Normalized fanout (each output) 3/: | |
| At low logic level | 10 maximum |
| At high logic level | 20 maximum 4/ |
| Case operating temperature range ($T_C$) | -55°C to +125°C |

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and Standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-38535 - Integrated Circuits (Microcircuits) Manufacturing, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-883 - Test Method Standard for Microelectronics.
MIL-STD-1835 - Interface Standard Electronic Component Case Outlines.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

1/ Must withstand the added $P_D$ due to short circuit test (e.g. $I_{OS}$).
2/ Maximum junction temperature shall not be exceeded except for short duration burn-in screening conditions per method 5004 of MIL-STD-883.
3/ The device will fanout in both high and low levels to the specified number of inputs on the same device type as that being tested.
4/ A fanout of 20 normalized loads is provided to facilitate connection of unused inputs to used inputs.



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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this document shall takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Microcircuits furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.3 and 6.4).

3.2 Item requirements. The individual item requirements shall be in accordance with MIL-PRF-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not affect the form, fit, or function as described herein.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-PRF-38535 and herein.

3.3.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.3.2 Logic diagrams. The logic diagrams shall be as specified on figure 2.

3.3.3 Truth tables. The truth tables shall be as specified on figure 3.

3.3.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.4 Lead material and finish. The lead material and finish shall be in accordance with MIL-PRF-38535 (see 6.6).

3.5 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.6 Electrical test requirements. Electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.7 Marking. Marking shall be in accordance with MIL-PRF-38535.

TABLE I. Electrical performance characteristics.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| High level output voltage | $V_{OH}$ | $V_{CC}$ = 4.5 V, $I_{OH}$ = -0.8 mA | 01,02, 04,05, 06 | 2.4 | | V |
| High level output leakage current | $I_{CEX}$ | $V_{CC}$ = 4.5 V, $V_{OUT}$ = 5.5 V | 03 | | 250 | µA |
| Low level output voltage | $V_{OL}$ | $V_{CC}$ = 4.5 V, $I_{OL}$ = 16 mA | All | | 0.4 | V |
| Input clamp voltage | $V_{IC}$ | $V_{CC}$ = 4.5 V, $I_{IN}$ = -12mA | All | | -1.5 | V |
| Low level input current | $I_{IL}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 0.4 V | 01,02, 03,06 | -0.7 | -1.6 | mA |
| | | | 04,05 | -0.4 | -1.3 | mA |
| High level input current | $I_{IH1}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 2.4 V | All | | 60 | µA |
| High level input current | $I_{IH2}$ | $V_{CC}$ = 5.5 V, $V_{IN}$ = 5.5 V | All | | 150 | µA |
| Short circuit output current | $I_{OS}$ | $V_{CC}$ = 5.5 V, $V_{OUT}$ = 0 V 1/ | 02 | -20 | -55 | mA |
| | | | 04,05,06 | -10 | -55 | mA |
| | | | 01 | -20 | -70 | mA |

See footnote at end of table.



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TABLE I. Electrical performance characteristics – Continued.

| Test | Symbol | Conditions<br>-55°C ≤ $T_C$ ≤ +125°C | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Supply current | $I_{CC}$ | $V_{CC}$ = 5.5 V | 01 | | 62 | mA |
| | | | 02,03 | | 35 | mA |
| | | | 04 | | 25 | mA |
| | | | 05 | | 34 | mA |
| | | | 06 | | 44 | mA |
| Propagation delay time high-to-low level output from A, B, C, or D to output | $t_{PHL1}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 4) | 01 | 5 | 49 | ns |
| Propagation delay time low-to-high level output from A, B, C, or D to output | $t_{PLH1}$ | | 01 | 5 | 53 | ns |
| Propagation delay time high-to-low level output from G1 or G2 to output | $t_{PHL2}$ | | 01 | 5 | 42 | ns |
| Propagation delay time low-to-high level output from G1 or G2 to output | $t_{PLH2}$ | | 01 | 5 | 45 | ns |
| Propagation delay time low-to-high level output from 1G or 2G to output | $t_{PLH1}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 5) | 02 | 3 | 33 | ns |
| Propagation delay time high-to-low level output from 1G or 2G to output | $t_{PHL1}$ | | 02 | 3 | 42 | ns |
| Propagation delay time low-to-high level output A or B to output through two levels | $t_{PLH2}$ | | 02 | 3 | 33 | ns |
| Propagation delay time high-to-low level output A or B to output through two levels | $t_{PHL2}$ | | 02 | 3 | 42 | ns |
| Propagation delay time low-to-high level output 2C to output | $t_{PLH3}$ | | 02 | 3 | 33 | ns |
| Propagation delay time high-to-low level output 2C to output | $t_{PHL3}$ | | 02 | 3 | 42 | ns |

See footnote at end of table.



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TABLE I. Electrical performance characteristics – Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Propagation delay time low-to-high level output 1C to output | $t_{PLH4}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 5) | 02 | 3 | 38 | ns |
| Propagation delay time high-to-low level output 1C to output | $t_{PHL4}$ | | 02 | 3 | 46 | ns |
| Propagation delay time low-to-high level output A or B to output | $t_{PLH5}$ | | 02 | 3 | 48 | ns |
| Propagation delay time high-to-low level output A or B to output | $t_{PHL5}$ | | 02 | 3 | 48 | ns |
| Propagation delay time low-to-high level output from 1G or 2G to output | $t_{PLH1}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 6) | 03 | 3 | 43 | ns |
| Propagation delay time high-to-low level output from 1G or 2G to output | $t_{PHL1}$ | | 03 | 3 | 52 | ns |
| Propagation delay time low-to-high level output A or B to output through two levels | $t_{PLH2}$ | | 03 | 3 | 43 | ns |
| Propagation delay time high-to-low level output A or B to output through two levels | $t_{PHL2}$ | | 03 | 3 | 52 | ns |
| Propagation delay time low-to-high level output 2C to output | $t_{PLH3}$ | | 03 | 3 | 43 | ns |
| Propagation delay time high-to-low level output 2C to output | $t_{PHL3}$ | | 03 | 3 | 52 | ns |
| Propagation delay time low-to-high level output 1C to output | $t_{PLH4}$ | | 03 | 3 | 48 | ns |
| Propagation delay time high-to-low level output 1C to output | $t_{PHL4}$ | | 03 | 3 | 56 | ns |
| Propagation delay time low-to-high level output A or B to output | $t_{PLH5}$ | | 03 | 3 | 57 | ns |

See footnote at end of table.



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TABLE I. Electrical performance characteristics – Continued.

| Test | Symbol | Conditions<br>$-55°C \le T_C \le +125°C$ | Device type | Limits | | Unit |
|---|---|---|---|---|---|---|
| | | | | Min | Max | |
| Propagation delay time high-to-low level output A or B to output | $t_{PHL5}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 6) | 03 | 3 | 57 | ns |
| Propagation delay time high-to-low level output A, B, C, or D to output through two levels | $t_{PHL1}$ | $R_L$ = 390 Ω ± 5%<br>$C_L$ = 50 pF minimum ± 10%<br>(figure 7) | 04,05, 06 | 5 | 48 | ns |
| Propagation delay time low-to-high level output A, B, C, or D to output through two levels | $t_{PLH1}$ | | 04,05, 06 | 5 | 48 | ns |
| Propagation delay time high-to-low level output A, B, C, or D to output | $t_{PHL2}$ | | 04,05, 06 | 5 | 53 | ns |
| Propagation delay time low-to-high level output A, B, C, or D to output | $t_{PLH2}$ | | 04,05, 06 | 5 | 53 | ns |

1/ Not more than one output should be shorted at one time.

TABLE II. Electrical test requirements.

| MIL-PRF-38535 test requirements | Subgroups (see table III) | |
|---|---|---|
| | Class S devices | Class B devices |
| Interim electrical parameters | 1 | 1 |
| Final electrical test parameters | 1*, 2, 3, 7, 9, 10, 11 | 1*, 2, 3, 7, 9 |
| Group A test requirements | 1, 2, 3, 7, 8, 9, 10, 11 | 1, 2, 3, 7, 9, 10, 11 |
| Group B electrical test parameters when using the method 5005 QCI option | 1,2,3,7,8,9, 10,11 | N/A |
| Group C end-point electrical parameters | 1,2,3,7,8,9, 10,11 | 1, 2, 3 |
| Group D end-point electrical parameters | 1, 2, 3 | 1, 2, 3 |

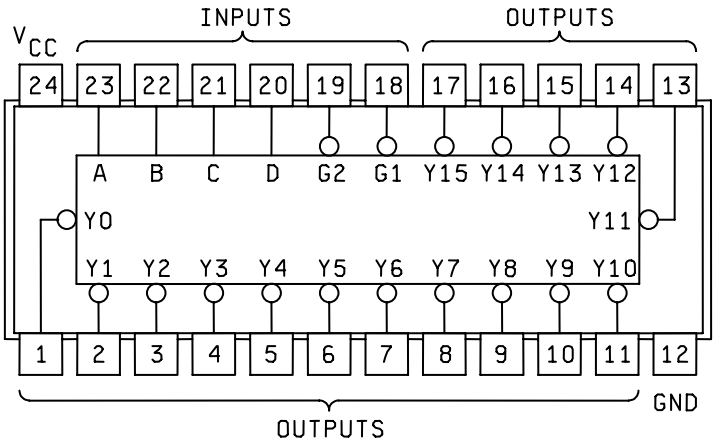
*PDA applies to subgroup 1.



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Device type 01

Cases J, K, and Z




Device types 02 and 03

Cases E and F

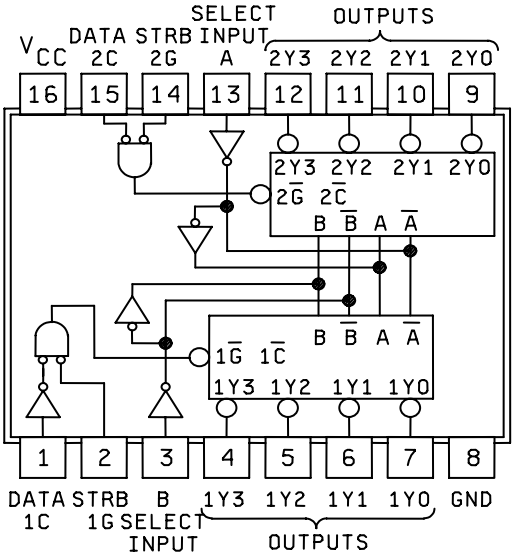


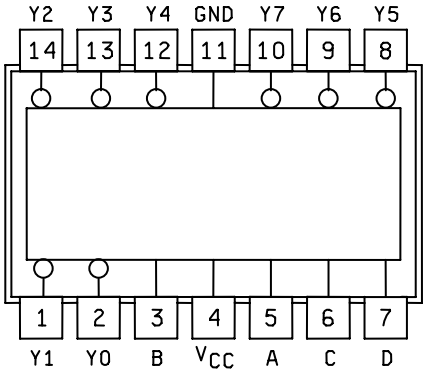

Figure 1. Terminal connections (top view).



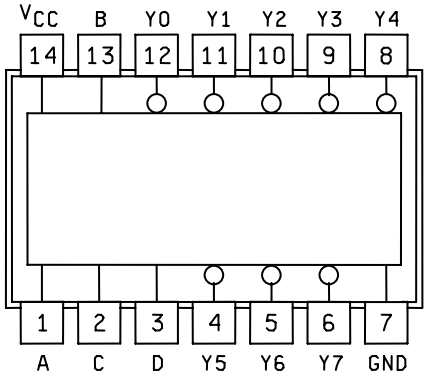
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Device type 04

CASES A,B AND D




CASE C




Device type 05 and 06

CASES E AND F

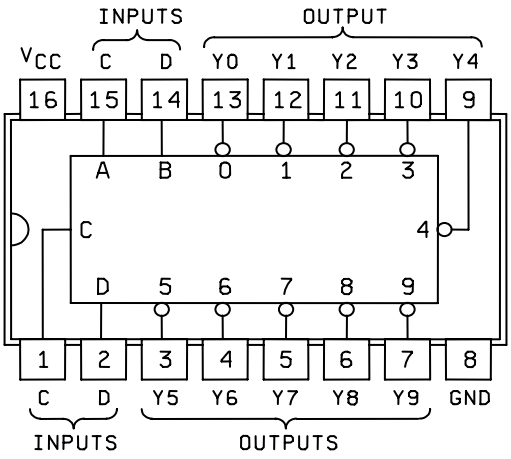



Figure 1. Terminal connections (top view) – Continued.



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Device type 01

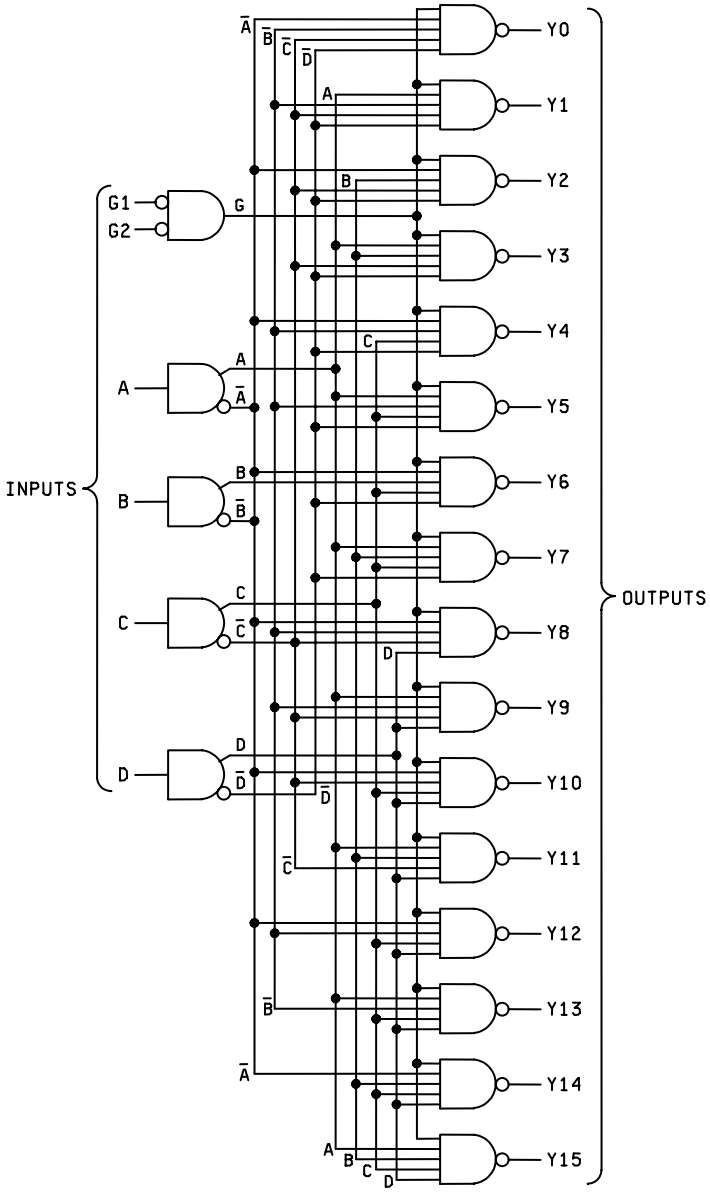



Figure 2. Logic diagrams.



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Device type 02 and 03

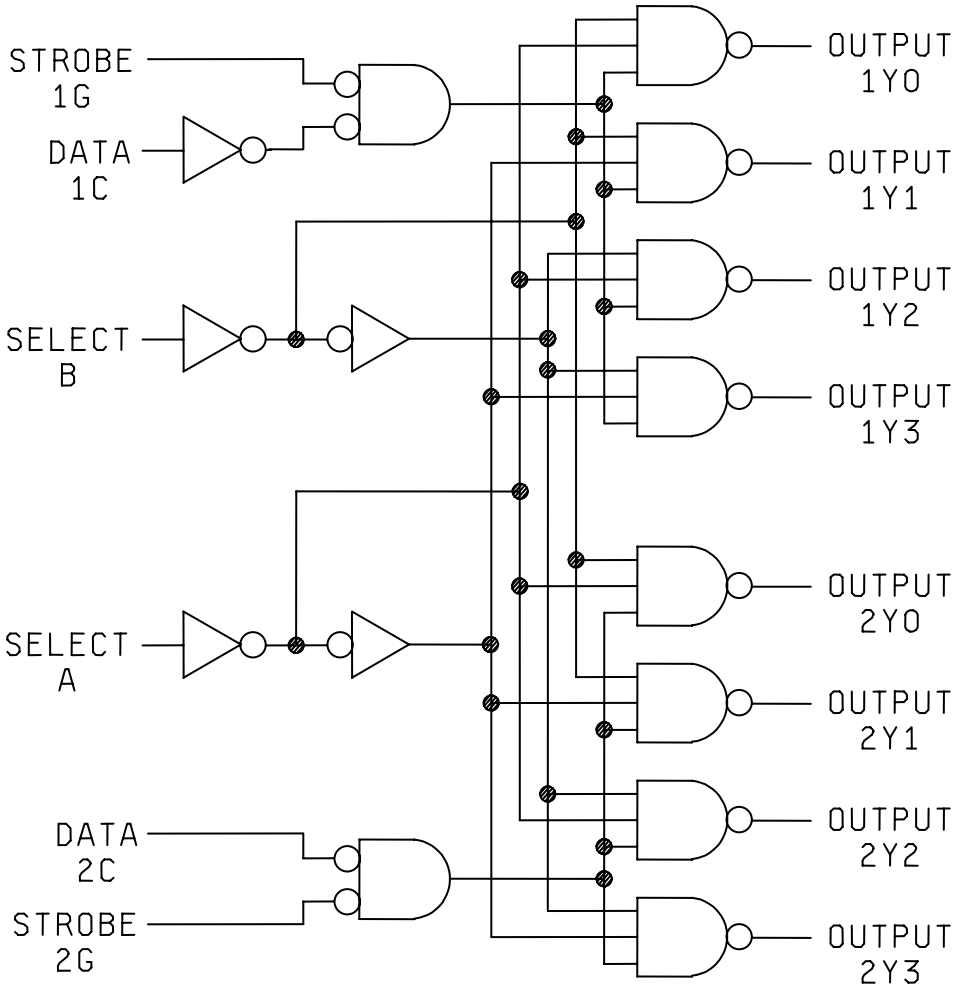
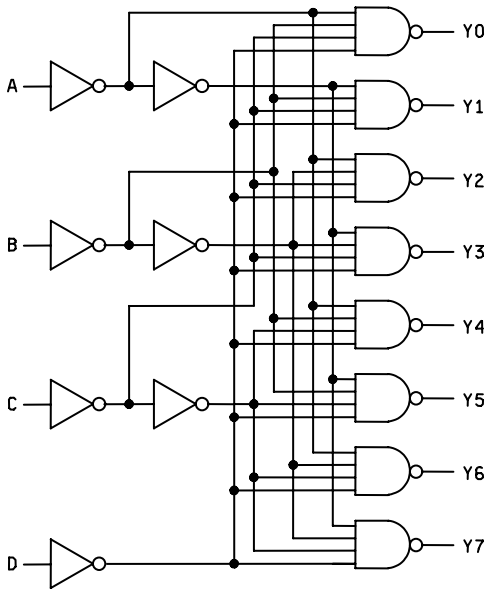



Figure 2. Logic diagrams – Continued.



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Device type 04




Device type 05

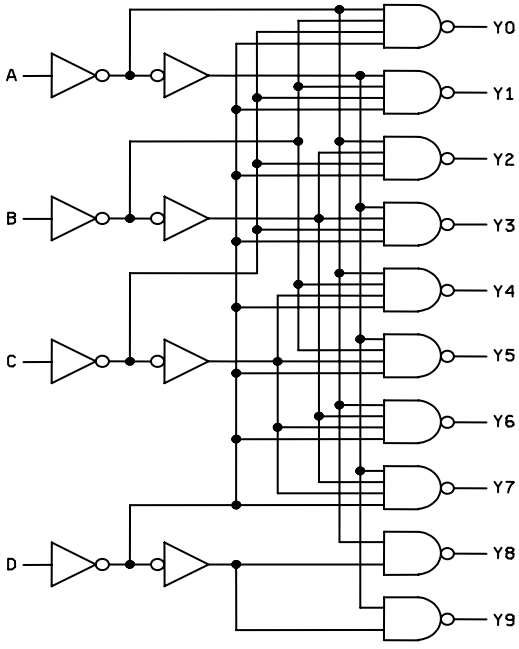



Figure 2. Logic diagrams – Continued.



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Device type 06

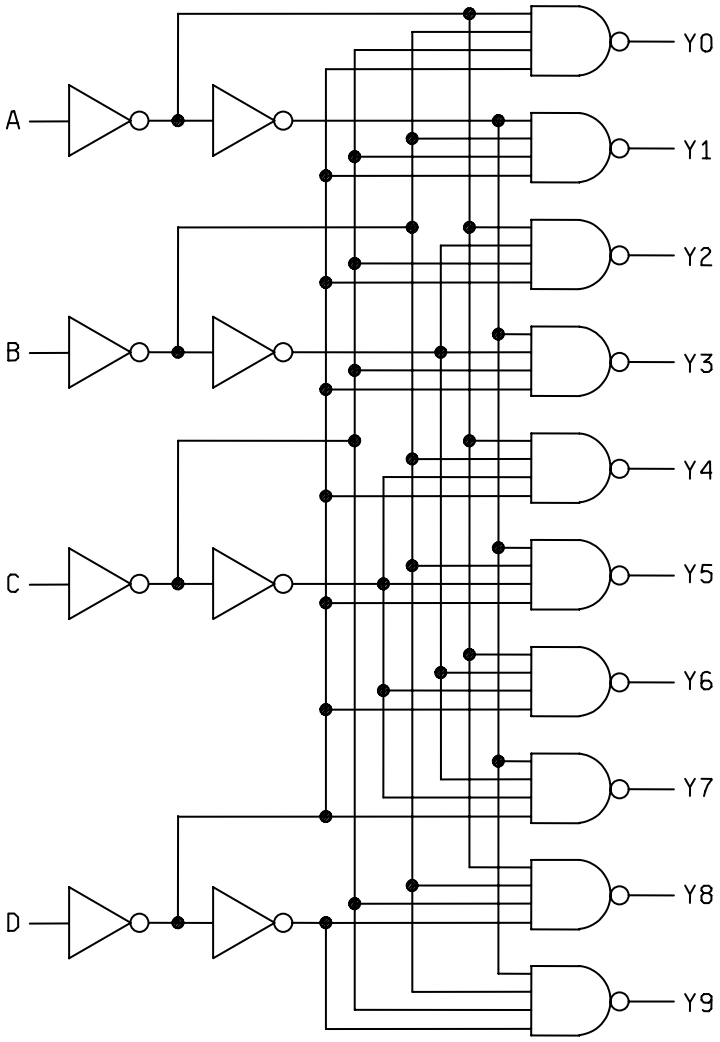



Figure 2. Logic diagrams – Continued.



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Device type 01

| Inputs | | | | | | Outputs | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| G1 | G2 | D | C | B | A | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 | Y11 | Y12 | Y13 | Y14 | Y15 |
| L | L | L | L | L | L | L | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H |
| L | L | L | L | L | H | H | L | H | H | H | H | H | H | H | H | H | H | H | H | H | H |
| L | L | L | L | H | L | H | H | L | H | H | H | H | H | H | H | H | H | H | H | H | H |
| L | L | L | L | H | H | H | H | H | L | H | H | H | H | H | H | H | H | H | H | H | H |
| L | L | L | H | L | L | H | H | H | H | L | H | H | H | H | H | H | H | H | H | H | H |
| L | L | L | H | L | H | H | H | H | H | H | L | H | H | H | H | H | H | H | H | H | H |
| L | L | L | H | H | L | H | H | H | H | H | H | L | H | H | H | H | H | H | H | H | H |
| L | L | L | H | H | H | H | H | H | H | H | H | H | L | H | H | H | H | H | H | H | H |
| L | L | H | L | L | L | H | H | H | H | H | H | H | H | L | H | H | H | H | H | H | H |
| L | L | H | L | L | H | H | H | H | H | H | H | H | H | H | L | H | H | H | H | H | H |
| L | L | H | L | H | L | H | H | H | H | H | H | H | H | H | H | L | H | H | H | H | H |
| L | L | H | L | H | H | H | H | H | H | H | H | H | H | H | H | H | L | H | H | H | H |
| L | L | H | H | L | L | H | H | H | H | H | H | H | H | H | H | H | H | L | H | H | H |
| L | L | H | H | L | H | H | H | H | H | H | H | H | H | H | H | H | H | H | L | H | H |
| L | L | H | H | H | L | H | H | H | H | H | H | H | H | H | H | H | H | H | H | L | H |
| L | L | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | L |
| L | H | X | X | X | X | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H |
| H | L | X | X | X | X | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H |
| H | H | X | X | X | X | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H | H |

H = High, L = Low, X = Irrelevant

Device types 02 and 03

2-LINE TO 4-LINE DECODER OR 1-LINE TO 4-LINE DEMULTIPLEXER

| Inputs | | | | Outputs | | | | | Inputs | | | | Outputs | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Select | | Strobe | Data | | | | | | Select | | Strobe | Data | | | | |
| B | A | 1G | 1C | 1Y0 | 1Y1 | 1Y2 | 1Y3 | | B | A | 2G | 2C | 2Y0 | 2Y1 | 2Y2 | 2Y3 |
| X | X | H | X | H | H | H | H | | X | X | H | X | H | H | H | H |
| L | L | L | H | L | H | H | H | | L | L | L | L | L | H | H | H |
| L | H | L | H | H | L | H | H | | L | H | L | L | H | L | H | H |
| H | L | L | H | H | H | L | H | | H | L | L | L | H | H | L | H |
| H | H | L | H | H | H | H | L | | H | H | L | L | H | H | H | L |
| X | X | X | L | H | H | H | H | | X | X | X | H | H | H | H | H |

FIGURE 3. Truth tables.



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Device types 02 and 03

3-LINE TO 8-LINE DECODER OR 1-LINE TO 8-LINE DEMULTIPLEXER

| Inputs | | | | Outputs | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Select | | | Strobe or Data | (0) | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| C* | B | A | G* | 2Y0 | 2Y1 | 2Y2 | 2Y3 | 1Y0 | 1Y1 | 1Y2 | 1Y3 |
| X | X | X | H | H | H | H | H | H | H | H | H |
| L | L | L | L | L | H | H | H | H | H | H | H |
| L | L | H | L | H | L | H | H | H | H | H | H |
| L | H | L | L | H | H | L | H | H | H | H | H |
| L | H | H | L | H | H | H | L | H | H | H | H |
| H | L | L | L | H | H | H | H | L | H | H | H |
| H | L | H | L | H | H | H | H | H | L | H | H |
| H | H | L | L | H | H | H | H | H | H | L | H |
| H | H | H | L | H | H | H | H | H | H | H | L |

C* = Inputs 1C and 2C connected together
G* = Inputs 1G and 2G connected together
H = High level, L = Low level, X = Irrelevant

Device type 04

| Inputs | | | | Outputs | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 |
| L | L | L | L | L | H | H | H | H | H | H | H |
| H | L | L | L | H | L | H | H | H | H | H | H |
| L | H | L | L | H | H | L | H | H | H | H | H |
| H | H | L | L | H | H | H | L | H | H | H | H |
| L | L | H | L | H | H | H | H | L | H | H | H |
| H | L | H | L | H | H | H | H | H | L | H | H |
| L | H | H | L | H | H | H | H | H | H | L | H |
| H | H | H | L | H | H | H | H | H | H | H | L |
| L | L | L | H | H | H | H | H | H | H | H | H |
| H | L | L | H | H | H | H | H | H | H | H | H |
| L | H | L | H | H | H | H | H | H | H | H | H |
| H | H | L | H | H | H | H | H | H | H | H | H |
| L | L | H | H | H | H | H | H | H | H | H | H |
| H | L | H | H | H | H | H | H | H | H | H | H |
| L | H | H | H | H | H | H | H | H | H | H | H |
| H | H | H | H | H | H | H | H | H | H | H | H |

H = High voltage level
L = Low voltage level

FIGURE 3. Truth tables – Continued.



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Device type 05

| Inputs | | | | Outputs | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 |
| L | L | L | L | L | H | H | H | H | H | H | H | H | H |
| H | L | L | L | H | L | H | H | H | H | H | H | H | H |
| L | H | L | L | H | H | L | H | H | H | H | H | H | H |
| H | H | L | L | H | H | H | L | H | H | H | H | H | H |
| L | L | H | L | H | H | H | H | L | H | H | H | H | H |
| H | L | H | L | H | H | H | H | H | L | H | H | H | H |
| L | H | H | L | H | H | H | H | H | H | L | H | H | H |
| H | H | H | L | H | H | H | H | H | H | H | L | H | H |
| L | L | L | H | H | H | H | H | H | H | H | H | L | H |
| H | L | L | H | H | H | H | H | H | H | H | H | H | L |
| L | H | L | H | H | H | H | H | H | H | H | H | L | H |
| H | H | L | H | H | H | H | H | H | H | H | H | H | L |
| L | L | H | H | H | H | H | H | H | H | H | H | L | H |
| H | L | H | H | H | H | H | H | H | H | H | H | H | L |
| L | H | H | H | H | H | H | H | H | H | H | H | L | H |
| H | H | H | H | H | H | H | H | H | H | H | H | H | L |

H = High voltage level
L = Low voltage level

Device type 06

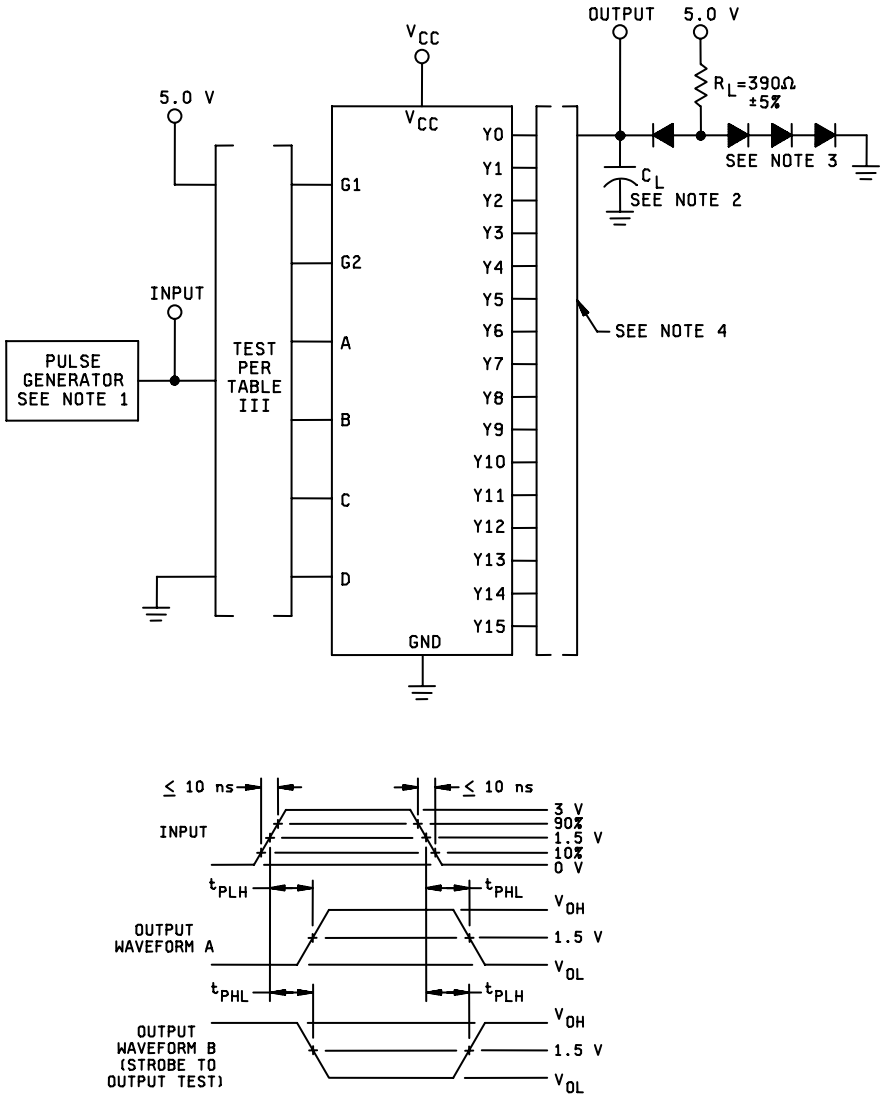
| Inputs | | | | Outputs | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 |
| L | L | L | L | L | H | H | H | H | H | H | H | H | H |
| H | L | L | L | H | L | H | H | H | H | H | H | H | H |
| L | H | L | L | H | H | L | H | H | H | H | H | H | H |
| H | H | L | L | H | H | H | L | H | H | H | H | H | H |
| L | L | H | L | H | H | H | H | L | H | H | H | H | H |
| H | L | H | L | H | H | H | H | H | L | H | H | H | H |
| L | H | H | L | H | H | H | H | H | H | L | H | H | H |
| H | H | H | L | H | H | H | H | H | H | H | L | H | H |
| L | L | L | H | H | H | H | H | H | H | H | H | L | H |
| H | L | L | H | H | H | H | H | H | H | H | H | H | L |
| L | H | L | H | H | H | H | H | H | H | H | H | H | H |
| H | H | L | H | H | H | H | H | H | H | H | H | H | H |
| L | L | H | H | H | H | H | H | H | H | H | H | H | H |
| H | L | H | H | H | H | H | H | H | H | H | H | H | H |
| L | H | H | H | H | H | H | H | H | H | H | H | H | H |
| H | H | H | H | H | H | H | H | H | H | H | H | H | H |

H = High voltage level
L = Low voltage level

FIGURE 3. Truth tables – Continued.



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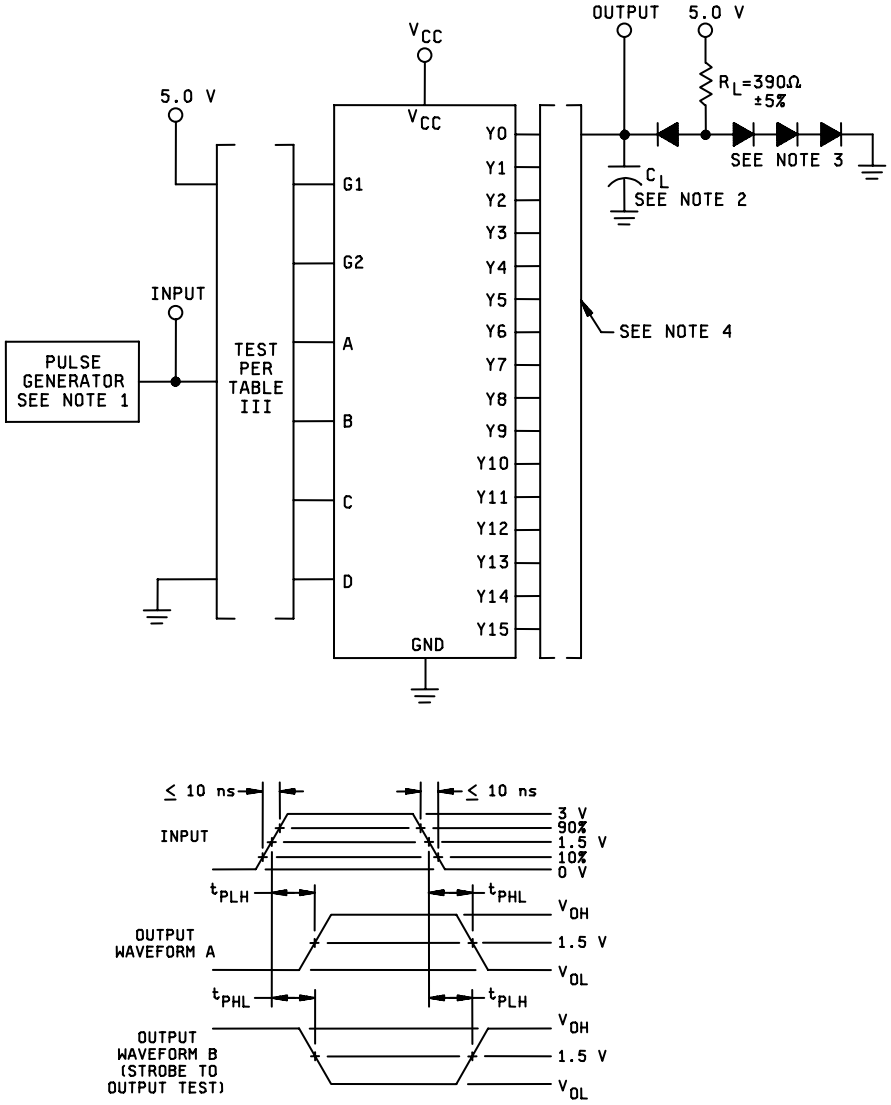

NOTES:

1. The pulse generator has the following characteristics: PRR ≤ 1 MHz, duty cycle = 50%, and $Z_{OUT}$ ≈ 50 Ω.
2. $C_L$ = 50 pF ± 10% minimum and includes probe and jig capacitance.
3. All diodes are 1N3064 or equivalent.
4. Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 4. Switching test for device type 01.



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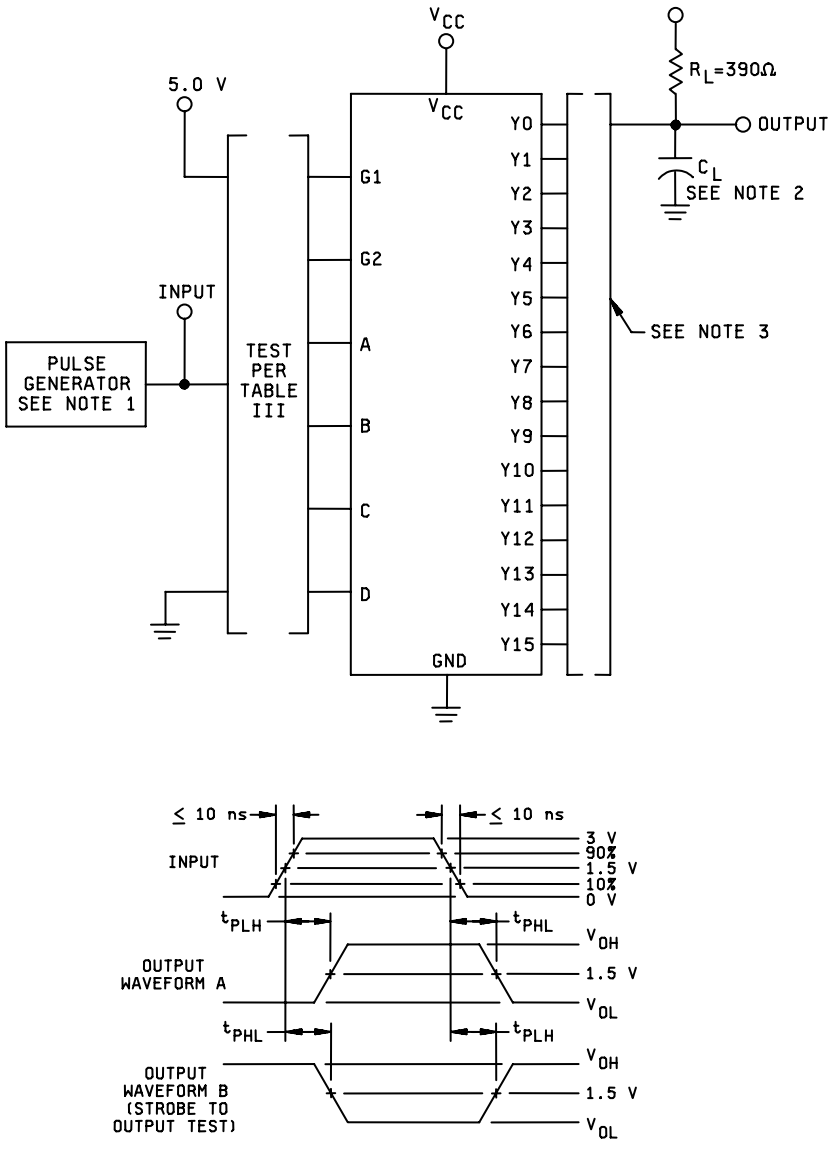

NOTES:

1 The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50%, and $Z_{OUT} \approx 50\ \Omega$.
2 $C_L$ = 50 pF minimum and includes probe and jig capacitance.
3 All diodes are 1N3064 or equivalent.
4 Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 5. Switching test for device type 02.



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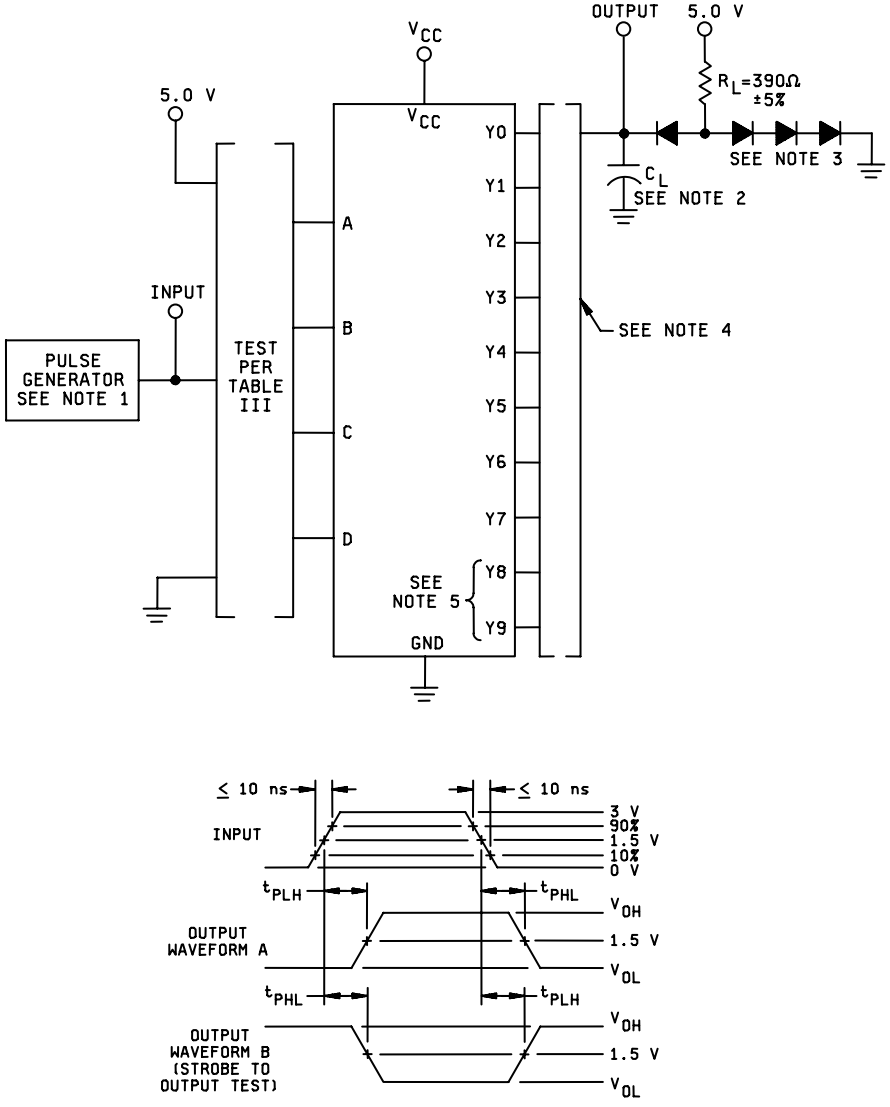

NOTES:

1 The pulse generator has the following characteristics: PRR = 1 MHz, duty cycle = 50%, and $Z_{OUT} \approx 50\ \Omega$.
2 $C_L$ = 15 pF minimum and includes probe and jig capacitance.
3 Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.

FIGURE 6. Switching test for device type 03.



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NOTES:

1 The pulse generator has the following characteristics: PRR ≤ 1 MHz, duty cycle = 50%, and $Z_{OUT}$ ≈ 50 Ω.
2 $C_L$ = 50 pF minimum and includes probe and jig capacitance.
3 All diodes are 1N3064 or equivalent.
4 Load circuit is applied separately to each output pin under test. All other outputs may be open or loaded.
5 Output pins Y8 and Y9 apply to device types 05 and 06 only.

FIGURE 7. Switching test for device types 04, 05, and 06 – Continued.



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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases J,K,Z | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 | GND | Y11 | Y12 | Y13 | Y14 | Y15 | G1 | G2 | D | C | B | A | $V_{CC}$ | | Min | Max | |
| 1<br>$T_C$=+25°C | $V_{OH}$ | 3006 | 1 | -.8mA | | | | | | | | | | | GND | | | | | | 2.0V | 2.0V | | | | | 4.5V | Y0 | 2.4 | | V |
| | | | 2 | | -.8mA | | | | | | | | | | " | | | | | | " | " | | | | | " | Y1 | " | | " |
| | | | 3 | | | -.8mA | | | | | | | | | " | | | | | | " | " | | | | | " | Y2 | " | | " |
| | | | 4 | | | | -.8mA | | | | | | | | " | | | | | | " | " | | | | | " | Y3 | " | | " |
| | | | 5 | | | | | -.8mA | | | | | | | " | | | | | | " | " | | | | | " | Y4 | " | | " |
| | | | 6 | | | | | | -.8mA | | | | | | " | | | | | | " | " | | | | | " | Y5 | " | | " |
| | | | 7 | | | | | | | -.8mA | | | | | " | | | | | | " | " | | | | | " | Y6 | " | | " |
| | | | 8 | | | | | | | | -.8mA | | | | " | | | | | | " | " | | | | | " | Y7 | " | | " |
| | | | 9 | | | | | | | | | -.8mA | | | " | | | | | | " | " | | | | | " | Y8 | " | | " |
| | | | 10 | | | | | | | | | | -.8mA | | " | | | | | | " | " | | | | | " | Y9 | " | | " |
| | | | 11 | | | | | | | | | | | -.8mA | " | | | | | | " | " | | | | | " | Y10 | " | | " |
| | | | 12 | | | | | | | | | | | | " | -.8mA | | | | | " | " | | | | | " | Y11 | " | | " |
| | | | 13 | | | | | | | | | | | | " | | -.8mA | | | | " | " | | | | | " | Y12 | " | | " |
| | | | 14 | | | | | | | | | | | | " | | | -.8mA | | | " | " | | | | | " | Y13 | " | | " |
| | | | 15 | | | | | | | | | | | | " | | | | -.8mA | | " | " | | | | | " | Y14 | " | | " |
| | | | 16 | | | | | | | | | | | | " | | | | | -.8mA | " | " | | | | | " | Y15 | " | | " |
| | $V_{OL}$ | 3007 | 17 | 16mA | | | | | | | | | | | " | | | | | | 0.8V | 0.8V | 0.8V | 0.8V | 0.8V | 0.8V | " | Y0 | | 0.4 | " |
| | | | 18 | | 16mA | | | | | | | | | | " | | | | | | 0.8V | " | " | " | 2.0V | 2.0V | " | Y1 | | " | " |
| | | | 19 | | | 16mA | | | | | | | | | " | | | | | | 2.0V | " | " | " | 2.0V | 0.8V | " | Y2 | | " | " |
| | | | 20 | | | | 16mA | | | | | | | | " | | | | | | 0.8V | " | " | " | 2.0V | 2.0V | " | Y3 | | " | " |
| | | | 21 | | | | | 16mA | | | | | | | " | | | | | | 0.8V | " | " | 2.0V | 0.8V | 0.8V | " | Y4 | | " | " |
| | | | 22 | | | | | | 16mA | | | | | | " | | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y5 | | " | " |
| | | | 23 | | | | | | | 16mA | | | | | " | | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y6 | | " | " |
| | | | 24 | | | | | | | | 16mA | | | | " | | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y7 | | " | " |
| | | | 25 | | | | | | | | | 16mA | | | " | | | | | | 0.8V | " | 2.0V | " | 0.8V | 0.8V | " | Y8 | | " | " |
| | | | 26 | | | | | | | | | | 16mA | | " | | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y9 | | " | " |
| | | | 27 | | | | | | | | | | | 16mA | " | | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y10 | | " | " |
| | | | 28 | | | | | | | | | | | | " | 16mA | | | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y11 | | " | " |
| | | | 29 | | | | | | | | | | | | " | | 16mA | | | | 0.8V | " | " | 2.0V | 0.8V | 0.8V | " | Y12 | | " | " |
| | | | 30 | | | | | | | | | | | | " | | | 16mA | | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y13 | | " | " |
| | | | 31 | | | | | | | | | | | | " | | | | 16mA | | 0.8V | " | " | " | 2.0V | 0.8V | " | Y14 | | " | " |
| | | | 32 | | | | | | | | | | | | " | | | | | 16mA | 0.8V | " | " | 2.0V | 2.0V | 0.8V | " | Y15 | | " | " |
| | $V_{IC}$ | | 33 | | | | | | | | | | | | " | | | | | | -12mA | | | | | | " | G1 | | -1.5 | " |
| | | | 34 | | | | | | | | | | | | " | | | | | | | -12mA | | | | | " | G2 | | " | " |
| | | | 35 | | | | | | | | | | | | " | | | | | | | " | | | | -12mA | " | A | | " | " |
| | | | 36 | | | | | | | | | | | | " | | | | | | | " | | | -12mA | | " | B | | " | " |
| | | | 37 | | | | | | | | | | | | " | | | | | | | " | | -12mA | | | " | C | | " | " |
| | | | 38 | | | | | | | | | | | | " | | | | | | | " | -12mA | | | | " | D | | " | " |
| | $I_{IL}$ | 3009 | 39 | | | | | | | | | | | | " | | | | | | 0.4V | 5.5V | | | | | 5.5V | G1 | -0.7 | -1.6 | mA |
| | | | 40 | | | | | | | | | | | | " | | | | | | 5.5V | 0.4V | | | | | " | G2 | " | " | " |
| | | | 41 | | | | | | | | | | | | " | | | | | | | | | | | 0.4V | " | A | | " | " |
| | | | 42 | | | | | | | | | | | | " | | | | | | | | | | 0.4V | | " | B | | " | " |
| | | | 43 | | | | | | | | | | | | " | | | | | | | | | 0.4V | | | " | C | | " | " |
| | | | 44 | | | | | | | | | | | | " | | | | | | | | 0.4V | | | | " | D | | " | " |
| | $I_{IH1}$ | 3010 | 45 | | | | | | | | | | | | " | | | | | | 2.4V | GND | | | | | " | G1 | | 60 | µA |
| | | | 46 | | | | | | | | | | | | " | | | | | | GND | 2.4V | | | | | " | G2 | | " | " |
| | | | 47 | | | | | | | | | | | | " | | | | | | | | | | | 2.4V | " | A | | " | " |
| | | | 48 | | | | | | | | | | | | " | | | | | | | | | | 2.4V | | " | B | | " | " |
| | | | 49 | | | | | | | | | | | | " | | | | | | | | | 2.4V | | | " | C | | " | " |
| | | | 50 | | | | | | | | | | | | " | | | | | | | | 2.4V | | | | " | D | | " | " |

See footnote at end of table.



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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases J,K,Z | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 | GND | Y11 | Y12 | Y13 | Y14 | Y15 | G1 | G2 | D | C | B | A | $V_{CC}$ | | Min | Max | |
| 1<br>$T_C$=+25°C | $I_{IH2}$ | 3010 | 51<br>52<br>53<br>54<br>55<br>56 | | | | | | | | | | | | GND<br>"<br>"<br>"<br>"<br>" | | | | | | 5.5V<br>GND | GND<br>5.5V | 5.5V | 5.5V | 5.5V | 5.5V<br>"<br>"<br>"<br>"<br>" | G1<br>G2<br>A<br>B<br>C<br>D | | 150<br>"<br>"<br>"<br>"<br>" | µA<br>"<br>"<br>"<br>"<br>" |
| | $I_{OS}$ | 3011 | 57<br>58<br>59<br>60<br>61<br>62<br>63<br>64<br>65<br>66<br>67<br>68<br>69<br>70<br>71<br>72 | GND | GND | GND | GND | GND | GND | GND | GND | GND | GND | GND | "<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | GND | GND | GND | GND | GND | 5.5V<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | 5.5V<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | | | | | "<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | Y0<br>Y1<br>Y2<br>Y3<br>Y4<br>Y5<br>Y6<br>Y7<br>Y8<br>Y9<br>Y10<br>Y11<br>Y12<br>Y13<br>Y14<br>Y15 | -20<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | -70<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | mA<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" |
| | $I_{CC}$ | 3005 | 73 | | | | | | | | | | | | " | | | | | | GND | GND | GND | GND | GND | GND | " | VCC | | 62 | " |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$=+125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$=-55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| 7<br>$T_C$=+25°C | Truth table tests | 3014 | 74<br>75<br>76<br>77<br>78<br>79<br>80<br>81<br>82<br>83<br>84<br>85<br>86<br>87<br>88<br>89<br>90<br>91<br>92 | L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>L<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>H<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | GND<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>"<br>" | H<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>L<br>H<br>"<br>"<br>" | B<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>A<br>" | B<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>A<br>" | B<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>X<br>" | B<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>X<br>" | B<br>B<br>A<br>A<br>B<br>B<br>A<br>A<br>B<br>B<br>A<br>A<br>B<br>B<br>A<br>A<br>X<br>" | B<br>A<br>B<br>A<br>B<br>A<br>B<br>A<br>B<br>A<br>B<br>A<br>B<br>A<br>B<br>A<br>X<br>" | 5.0V<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | 1/<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>"<br>" | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

See footnote at end of table.





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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases J,K,Z Test no. | 1 Y0 | 2 Y1 | 3 Y2 | 4 Y3 | 5 Y4 | 6 Y5 | 7 Y6 | 8 Y7 | 9 Y8 | 10 Y9 | 11 Y10 | 12 GND | 13 Y11 | 14 Y12 | 15 Y13 | 16 Y14 | 17 Y15 | 18 G1 | 19 G2 | 20 D | 21 C | 22 B | 23 A | 24 $V_{CC}$ | Measured terminal | Limits Min | Limits Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 9 $T_C$=+25°C | $t_{PHL1}$ | 3003 (Fig 4) | 93 | OUT | | | | | | | | | | | GND | | | | | | GND | GND | GND | GND | GND | IN | 5.0V | Y0 | 5 | 38 | ns |
| | | | 94 | | OUT | | | | | | | | | | " | | | | | | " | " | " | GND | IN | 5.0V | " | Y1 | " | " | " |
| | | | 95 | | | OUT | | | | | | | | | " | | | | | | " | " | " | IN | 5.0V | GND | " | Y2 | " | " | " |
| | | | 96 | | | | OUT | | | | | | | | " | | | | | | " | " | IN | GND | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 97 | | | | | OUT | | | | | | | " | | | | | | " | " | GND | 5.0V | GND | IN | " | Y4 | " | " | " |
| | | | 98 | | | | | | OUT | | | | | | " | | | | | | " | " | " | " | IN | 5.0V | " | Y5 | " | " | " |
| | | | 99 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | 5.0V | IN | " | Y6 | " | " | " |
| | | | 100 | | | | | | | | OUT | | | | " | | | | | | " | " | IN | " | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 101 | | | | | | | | | OUT | | | " | | | | | | " | " | 5.0V | GND | GND | IN | " | Y8 | " | " | " |
| | | | 102 | | | | | | | | | | OUT | | " | | | | | | " | " | " | GND | IN | 5.0V | " | Y9 | " | " | " |
| | | | 103 | | | | | | | | | | | OUT | " | | | | | | " | " | " | IN | 5.0V | GND | " | Y10 | " | " | " |
| | | | 104 | | | | | | | | | | | | " | OUT | | | | | " | " | " | IN | 5.0V | 5.0V | " | Y11 | " | " | " |
| | | | 105 | | | | | | | | | | | | " | | OUT | | | | " | " | " | 5.0V | GND | IN | " | Y12 | " | " | " |
| | $t_{PHL1}$ | 3011 (Fig 4) | 106 | | | | | | | | | | | | " | | | OUT | | | " | " | " | " | IN | 5.0V | " | Y13 | " | " | " |
| | | | 107 | | | | | | | | | | | | " | | | | OUT | | " | " | " | " | 5.0V | IN | " | Y14 | " | " | " |
| | $t_{PLH1}$ | | 108 | | OUT | | | | | | | | | | " | | | | | | " | " | GND | GND | GND | IN | " | Y1 | " | 41 | " |
| | | | 109 | | | OUT | | | | | | | | | " | | | | | | " | " | " | " | IN | GND | " | Y2 | " | " | " |
| | | | 110 | | | | OUT | | | | | | | | " | | | | | | " | " | " | " | IN | 5.0V | " | Y3 | " | " | " |
| | | | 111 | | | | | OUT | | | | | | | " | | | | | | " | " | " | IN | GND | GND | " | Y4 | " | " | " |
| | | | 112 | | | | | | OUT | | | | | | " | | | | | | " | " | " | " | GND | 5.0V | " | Y5 | " | " | " |
| | | | 113 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | 5.0V | GND | " | Y6 | " | " | " |
| | | | 114 | | | | | | | | OUT | | | | " | | | | | | " | " | " | " | 5.0V | IN | " | Y7 | " | " | " |
| | | | 115 | | | | | | | | | OUT | | | " | | | | | | " | " | IN | GND | GND | GND | " | Y8 | " | " | " |
| | | | 116 | | | | | | | | | | OUT | | " | | | | | | " | " | " | " | GND | 5.0V | " | Y9 | " | " | " |
| | | | 117 | | | | | | | | | | | OUT | " | | | | | | " | " | " | " | 5.0V | GND | " | Y10 | " | " | " |
| | | | 118 | | | | | | | | | | | | " | OUT | | | | | " | " | " | " | 5.0V | 5.0V | " | Y11 | " | " | " |
| | | | 119 | | | | | | | | | | | | " | | OUT | | | | " | " | " | 5.0V | GND | GND | " | Y12 | " | " | " |
| | | | 120 | | | | | | | | | | | | " | | | OUT | | | " | " | 5.0V | 5.0V | GND | IN | " | Y13 | " | " | " |
| | | | 121 | | | | | | | | | | | | " | | | | OUT | | " | " | 5.0V | IN | 5.0V | GND | " | Y14 | " | " | " |
| | | | 122 | | | | | | | | | | | | " | | | | | OUT | " | " | IN | 5.0V | 5.0V | 5.0V | " | Y15 | " | " | " |
| | $t_{PHL2}$ | | 123 | OUT | | | | | | | | | | | " | | | | | | IN | " | GND | GND | GND | GND | " | Y0 | " | 32 | " |
| | | | 124 | | | | | | | | | | | | " | | | | | OUT | GND | IN | 5.0V | 5.0V | 5.0V | 5.0V | " | Y15 | " | 32 | " |
| | $t_{PLH2}$ | | 125 | OUT | | | | | | | | | | | " | | | | | | IN | GND | GND | GND | GND | GND | " | Y0 | " | 35 | " |
| | | | 126 | | | | | | | | | | | | " | | | | | OUT | GND | IN | 5.0V | 5.0V | 5.0V | 5.0V | " | Y15 | " | " | " |
| | $t_{PHL3}$ | | 127 | | | | | | OUT | | | | | | " | | | | | | " | GND | GND | " | GND | IN | " | Y5 | " | " | " |
| | | | 128 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | IN | GND | " | Y6 | " | " | " |
| | | | 129 | | | | | | | | OUT | | | | " | | | | | | " | " | " | IN | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 130 | | | | | | | | | OUT | | | " | | | | | | " | " | IN | GND | GND | GND | " | Y8 | " | " | " |
| 10 $T_C$=+125°C | $t_{PHL1}$ | | 131 | OUT | | | | | | | | | | | " | | | | | | " | " | GND | " | GND | IN | " | Y0 | " | 49 | " |
| | | | 132 | | OUT | | | | | | | | | | " | | | | | | " | " | " | " | IN | 5.0V | " | Y1 | " | " | " |
| | | | 133 | | | OUT | | | | | | | | | " | | | | | | " | " | " | IN | 5.0V | GND | " | Y2 | " | " | " |
| | | | 134 | | | | OUT | | | | | | | | " | | | | | | " | " | IN | GND | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 135 | | | | | OUT | | | | | | | " | | | | | | " | " | GND | 5.0V | GND | IN | " | Y4 | " | " | " |
| | | | 136 | | | | | | OUT | | | | | | " | | | | | | " | " | " | " | IN | 5.0V | " | Y5 | " | " | " |
| | | | 137 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | 5.0V | IN | " | Y6 | " | " | " |
| | | | 138 | | | | | | | | OUT | | | | " | | | | | | " | " | IN | " | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 139 | | | | | | | | | OUT | | | " | | | | | | " | " | 5.0V | GND | GND | IN | " | Y8 | " | " | " |
| | | | 140 | | | | | | | | | | OUT | | " | | | | | | " | " | " | GND | IN | 5.0V | " | Y9 | " | " | " |
| | | | 141 | | | | | | | | | | | OUT | " | | | | | | " | " | " | IN | 5.0V | GND | " | Y10 | " | " | " |
| | | | 142 | | | | | | | | | | | | " | OUT | | | | | " | " | " | IN | 5.0V | 5.0V | " | Y11 | " | " | " |
| | | | 143 | | | | | | | | | | | | " | | OUT | | | | " | " | " | 5.0V | GND | IN | " | Y12 | " | " | " |
| | | | 144 | | | | | | | | | | | | " | | | OUT | | | " | " | " | " | IN | 5.0V | " | Y13 | " | " | " |
| | | | 145 | | | | | | | | | | | | " | | | | OUT | | " | " | " | " | 5.0V | IN | " | Y14 | " | " | " |

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TABLE III. Group A inspection for device type 01 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases J,K,Z | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | Y0 | Y1 | Y2 | Y3 | Y4 | Y5 | Y6 | Y7 | Y8 | Y9 | Y10 | GND | Y11 | Y12 | Y13 | Y14 | Y15 | G1 | G2 | D | C | B | A | $V_{CC}$ | | Min | Max | |
| 10 $T_C$=+125°C | $t_{PLH1}$ | 3003 (Fig 4) | 146 | | OUT | | | | | | | | | | GND | | | | | | GND | GND | GND | GND | GND | IN | 5.0V | Y1 | 5 | 53 | ns |
| | | | 147 | | | OUT | | | | | | | | | " | | | | | | " | " | " | " | IN | GND | " | Y2 | " | " | " |
| | | | 148 | | | | OUT | | | | | | | | " | | | | | | " | " | " | " | IN | 5.0V | " | Y3 | " | " | " |
| | | | 149 | | | | | OUT | | | | | | | " | | | | | | " | " | " | IN | GND | GND | " | Y4 | " | " | " |
| | | | 150 | | | | | | OUT | | | | | | " | | | | | | " | " | " | " | GND | 5.0V | " | Y5 | " | " | " |
| | | | 151 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | 5.0V | GND | " | Y6 | " | " | " |
| | | | 152 | | | | | | | | OUT | | | | " | | | | | | " | " | " | 5.0V | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 153 | | | | | | | | | OUT | | | " | | | | | | " | " | IN | GND | GND | GND | " | Y8 | " | " | " |
| | | | 154 | | | | | | | | | | OUT | | " | | | | | | " | " | " | " | GND | 5.0V | " | Y9 | " | " | " |
| | | | 155 | | | | | | | | | | | OUT | " | | | | | | " | " | " | " | 5.0V | GND | " | Y10 | " | " | " |
| | | | 156 | | | | | | | | | | | | " | OUT | | | | | " | " | " | " | 5.0V | 5.0V | " | Y11 | " | " | " |
| | | | 157 | | | | | | | | | | | | " | | OUT | | | | " | " | " | 5.0V | GND | GND | " | Y12 | " | " | " |
| | | | 158 | | | | | | | | | | | | " | | | OUT | | | " | " | 5.0V | 5.0V | GND | IN | " | Y13 | " | " | " |
| | | | 159 | | | | | | | | | | | | " | | | | OUT | | " | " | 5.0V | IN | 5.0V | GND | " | Y14 | " | " | " |
| | | | 160 | | | | | | | | | | | | " | | | | | OUT | " | " | IN | 5.0V | 5.0V | 5.0V | " | Y15 | " | " | " |
| | $t_{PHL2}$ | | 161 | OUT | | | | | | | | | | | " | | | | | | IN | " | GND | GND | GND | GND | " | Y0 | " | 42 | " |
| | | | 162 | | | | | | | | | | | | " | | | | | OUT | GND | IN | 5.0V | 5.0V | 5.0V | 5.0V | " | Y15 | " | 42 | " |
| | $t_{PLH2}$ | | 163 | OUT | | | | | | | | | | | " | | | | | | IN | GND | GND | GND | GND | GND | " | Y0 | " | 45 | " |
| | | | 164 | | | | | | | | | | | | " | | | | | OUT | GND | IN | 5.0V | 5.0V | 5.0V | 5.0V | " | Y15 | " | " | " |
| | | | 165 | | | | | | OUT | | | | | | " | | | | | | " | GND | GND | " | GND | IN | " | Y5 | " | " | " |
| | | | 166 | | | | | | | OUT | | | | | " | | | | | | " | " | " | " | IN | GND | " | Y6 | " | " | " |
| | | | 167 | | | | | | | | OUT | | | | " | | | | | | " | " | " | IN | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 168 | | | | | | | | | OUT | | | " | | | | | | " | " | IN | GND | GND | GND | " | Y8 | " | " | " |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | | | | | | | | | |

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 1 $T_C$=+25°C | $V_{OH}$ | 3006 | 1 | | 2.0V | | | | -.8mA | -.8mA | GND | | | | | | | | 4.5V | 1Y0 | 2.4 | | V |
| | | | 2 | | " | | | -.8mA | | | " | | | | | | | | " | 1Y1 | " | | " |
| | | | 3 | | " | | -.8mA | | | | " | -.8mA | | | | | | | " | 1Y2 | " | | " |
| | | | 4 | | | | | | | | " | | -.8mA | | | | | | " | 1Y3 | " | | " |
| | | | 5 | | | | | | | | " | | | -.8mA | | | 2.0V | | " | 2Y0 | " | | " |
| | | | 6 | | | | | | | | " | | | | | | " | | " | 2Y1 | " | | " |
| | | | 7 | | | | | | | | " | | | | -.8mA | | " | | " | 2Y2 | " | | " |
| | | | 8 | | | | | | | | " | | | | | | " | | " | 2Y3 | " | | " |
| | $V_{OL}$ | 3007 | 9 | 2.0V | 0.8V | 0.8V | | | | 16mA | " | | | | 16mA | 0.8V | | | " | 1Y0 | | 0.4 | " |
| | | | 10 | " | " | 0.8V | | | 16mA | | " | | | | | 2.0V | | | " | 1Y1 | | " | " |
| | | | 11 | " | " | 2.0V | | 16mA | | | " | | | | | 0.8V | | | " | 1Y2 | | " | " |
| | | | 12 | " | " | 2.0V | 16mA | | | | " | | | | | 2.0V | | | " | 1Y3 | | " | " |
| | | | 13 | | | 0.8V | | | | | " | 16mA | | | | 0.8V | 0.8V | | " | 2Y0 | | " | " |
| | | | 14 | | | 0.8V | | | | | " | | 16mA | | | 2.0V | " | | " | 2Y1 | | " | " |
| | | | 15 | | | 2.0V | | | | | " | | | 16mA | | 0.8V | " | | " | 2Y2 | | " | " |
| | | | 16 | | | 2.0V | | | | | " | | | | | 2.0V | " | | " | 2Y3 | | " | " |
| | $V_{IC}$ | | 17 | | | | | | | | " | | | | | -12mA | | | " | A | | -1.5 | " |
| | | | 18 | -12mA | | -12mA | | | | | " | | | | | | | | " | B | | " | " |
| | | | 19 | | -12mA | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 20 | | | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 21 | | | | | | | | " | | | | | | | -12mA | " | 2C | | " | " |
| | | | 22 | | | | | | | | " | | | | | | -12mA | | " | 2G | | " | " |
| | $I_{IL}$ | 3009 | 23 | | | | | | | | " | | | | | 0.4V | | | 5.5V | A | -0.7 | -1.6 | mA |
| | | | 24 | | | 0.4V | | | | | " | | | | | | | | " | B | " | " | " |
| | | | 25 | 0.4V | | | | | | | " | | | | | | | | " | 1C | " | " | " |
| | | | 26 | GND | 0.4V | | | | | | " | | | | | | | | " | 1G | " | " | " |
| | | | 27 | | | | | | | | " | | | | | | | 0.4V | " | 2C | " | " | " |
| | | | 28 | | | | | | | | " | | | | | | 0.4V | | " | 2G | " | " | " |
| | $I_{IH1}$ | 3010 | 29 | | | | | | | | " | | | | | 2.4V | | | " | A | | 60 | µA |
| | | | 30 | | | 2.4V | | | | | " | | | | | | | | " | B | | " | " |
| | | | 31 | 2.4V | | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 32 | 5.5V | 2.4V | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 33 | | | | | | | | " | | | | | | | 2.4V | " | 2C | | " | " |
| | | | 34 | | | | | | | | " | | | | | | 2.4V | | " | 2G | | " | " |
| | $I_{IH2}$ | | 35 | | | | | | | | " | | | | | 5.5V | | | " | A | | 150 | " |
| | | | 36 | | | 5.5V | | | | | " | | | | | | | | " | B | | " | " |
| | | | 37 | 5.5V | | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 38 | 5.5V | 5.5V | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 39 | | | | | | | | " | | | | | | | 5.5V | " | 2C | | " | " |
| | | | 40 | | | | | | | | " | | | | | | 5.5V | | " | 2G | | " | " |
| | $I_{OS}$ | 3011 | 41 | | 5.5V | | | | | | " | | | | | | | | " | 1Y0 | -20 | -55 | mA |
| | | | 42 | | " | | | GND | GND | GND | " | | | | | | | | " | 1Y1 | " | " | " |
| | | | 43 | | " | | | | | | " | | | | | | | | " | 1Y2 | " | " | " |
| | | | 44 | | | | GND | | | | " | GND | | | | | | | " | 1Y3 | " | " | " |
| | | | 45 | | | | | | | | " | | GND | | | | | | " | 2Y0 | " | " | " |
| | | | 46 | | | | | | | | " | | | GND | | | 5.5V | | " | 2Y1 | " | " | " |
| | | | 47 | | | | | | | | " | | | | | | " | | " | 2Y2 | " | " | " |
| | | | 48 | | | | | | | | " | | | | GND | | " | | " | 2Y3 | " | " | " |
| | $I_{CC}$ | 3005 | 49 | | | | | | | | " | | | | | | GND | GND | " | VCC | | 35 | " |

See footnote at end of table.





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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$=+25°C | Truth table test | 3014 | 50 | X | A | X | H | H | H | H | GND | H | H | H | H | X | A | X | 5.0V | 1/ | | | |
| | | | 51 | A | B | B | " | " | H | L | " | L | H | " | " | B | B | B | " | " | | | |
| | | | 52 | " | " | B | " | " | L | H | " | H | L | " | " | A | " | " | " | " | | | |
| | | | 53 | " | " | A | " | L | H | " | " | " | H | L | " | B | " | " | " | " | | | |
| | | | 54 | " | " | A | L | H | " | " | " | " | " | H | L | A | " | " | " | " | | | |
| | | | 55 | B | X | X | H | " | " | " | " | " | " | " | H | X | X | A | " | " | | | |
| | | | 56 | X | A | X | " | " | " | " | " | " | " | " | " | X | A | X | " | " | | | |
| | | | 57 | B | B | B | " | " | " | " | " | L | " | " | " | B | B | B | " | " | | | |
| | | | 58 | " | " | B | " | " | " | " | " | H | L | " | " | A | " | " | " | " | | | |
| | | | 59 | " | " | A | " | " | " | " | " | " | H | L | " | B | " | " | " | " | | | |
| | | | 60 | " | " | A | " | " | " | " | " | " | " | H | L | A | " | " | " | " | | | |
| | | | 61 | A | " | B | " | " | " | L | " | " | " | " | H | B | " | A | " | " | | | |
| | | | 62 | " | " | B | " | " | L | H | " | " | " | " | " | A | " | " | " | " | | | |
| | | | 63 | " | " | A | " | L | H | " | " | " | " | " | " | B | " | " | " | " | | | |
| | | | 64 | " | " | A | L | H | H | " | " | " | " | " | " | A | " | " | " | " | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$=+25°C | $t_{PLH1}$ | 3003 (Fig 5) | 65 | 5.0V | IN | GND | | | | OUT | GND | | | | | GND | | | 5.0V | 1Y0 | 3 | 25 | ns |
| | | | 66 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 67 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 68 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 69 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 70 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 71 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 72 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PHL1}$ | | 73 | 5.0V | IN | GND | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 32 | " |
| | | | 74 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 75 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 76 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 77 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 78 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 79 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 80 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PLH2}$ | | 81 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 25 | " |
| | | | 82 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 83 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 84 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PHL2}$ | | 85 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 32 | " |
| | | | 86 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 87 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 88 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PLH3}$ | | 89 | | | GND | | | | | " | OUT | | | | GND | GND | IN | " | 2Y0 | " | 25 | " |
| | $t_{PHL3}$ | | 90 | | | GND | | | | | " | OUT | | | | GND | GND | IN | " | 2Y0 | " | 32 | " |

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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 9 $T_C$=+25°C | $t_{PLH4}$ | 3003 (Fig 5) | 91 | IN | GND | GND | | | | OUT | GND | | | | | GND | | | 5.0V | 1Y0 | 3 | 29 | ns |
| | $t_{PHL4}$ | | 92 | IN | GND | GND | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 35 | " |
| | $t_{PLH5}$ | | 93 | 5.0V | GND | IN | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | 37 | " |
| | | | 94 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 95 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 96 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| | $t_{PHL5}$ | | 97 | 5.0V | GND | IN | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 98 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 99 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 100 | | | " | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| 10 $T_C$=+125°C | $t_{PLH1}$ | | 101 | 5.0V | IN | " | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 33 | " |
| | | | 102 | " | " | " | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 103 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 104 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 105 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 106 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 107 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 108 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PHL1}$ | | 109 | 5.0V | IN | GND | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 42 | " |
| | | | 110 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 111 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 112 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 113 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 114 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 115 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 116 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PLH2}$ | | 117 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 33 | " |
| | | | 118 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 119 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 120 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PHL2}$ | | 121 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 42 | " |
| | | | 122 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 123 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 124 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PLH3}$ | | 125 | | | GND | | | | | " | OUT | | | | GND | " | IN | " | 2Y0 | " | 33 | " |
| | $t_{PHL3}$ | | 126 | | | " | | | | | " | OUT | | | | " | " | IN | " | 2Y0 | " | 42 | " |
| | $t_{PLH4}$ | | 127 | IN | GND | " | | | | OUT | " | | | | | " | | | " | 1Y0 | " | 38 | " |
| | $t_{PHL4}$ | | 128 | IN | GND | " | | | | OUT | " | | | | | " | | | " | 1Y0 | " | 46 | " |

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TABLE III. Group A inspection for device type 02 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 10 $T_C$=+125°C | $t_{PLH5}$ | 3003 (Fig 4) | 129 | 5.0V | GND | IN | | OUT | | | GND | | | OUT | | GND | GND | GND | 5.0V | 1Y2 | 3 | 48 | ns |
| | | | 130 | | | IN | | | | | " | | | | | GND | | | " | 2Y2 | " | " | " |
| | | | 131 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 132 | | | GND | | | | | " | | | OUT | | IN | | | " | 2Y2 | " | " | " |
| | $t_{PHL5}$ | | 133 | 5.0V | GND | IN | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 134 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 135 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 136 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y2 | " | " | " |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 1 $T_C$=+25°C | $I_{CEX}$ | | 1 | | 2.0V | | | | 5.5V | 5.5V | GND | | | | | | | | 4.5V | 1Y0 | | 250 | µA |
| | | | 2 | | " | | | | | | " | | | | | | | | " | 1Y1 | | " | " |
| | | | 3 | | " | | | 5.5V | | | " | | | | | | | | " | 1Y2 | | " | " |
| | | | 4 | | " | | 5.5V | | | | " | | | | | | | | " | 1Y3 | | " | " |
| | | | 5 | | | | | | | | " | 5.5V | | | | | 2.0V | | " | 2Y0 | | " | " |
| | | | 6 | | | | | | | | " | | 5.5V | | | | " | | " | 2Y1 | | " | " |
| | | | 7 | | | | | | | | " | | | 5.5V | | | " | | " | 2Y2 | | " | " |
| | | | 8 | | | | | | | | " | | | | 5.5V | | | | " | 2Y3 | | " | " |
| | $V_{OL}$ | 3007 | 9 | 2.0V | 0.8V | 0.8V | | | | | " | | | | | 0.8V | | | " | 1Y0 | | 0.4 | V |
| | | | 10 | " | " | 0.8V | | | | | " | | | | | 2.0V | | | " | 1Y1 | | " | " |
| | | | 11 | " | " | 2.0V | | 16mA | | | " | | | | | 0.8V | | | " | 1Y2 | | " | " |
| | | | 12 | | | 2.0V | 16mA | | 16mA | | " | | | | | 2.0V | | | " | 1Y3 | | " | " |
| | | | 13 | | | 0.8V | | | | | " | 16mA | | | | 0.8V | 0.8V | 0.8V | " | 2Y0 | | " | " |
| | | | 14 | | | 0.8V | | | | | " | | 16mA | | | 2.0V | " | " | " | 2Y1 | | " | " |
| | | | 15 | | | 2.0V | | | | | " | | | 16mA | | 0.8V | " | " | " | 2Y2 | | " | " |
| | | | 16 | | | 2.0V | | | | | " | | | | 16mA | 2.0V | " | " | " | 2Y3 | | " | " |
| | $V_{IC}$ | | 17 | | | | | | | | " | | | | | -12mA | | | " | A | | -1.5 | " |
| | | | 18 | | | -12mA | | | | | " | | | | | | | | " | B | | " | " |
| | | | 19 | -12mA | | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 20 | | -12mA | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 21 | | | | | | | | " | | | | | | | -12mA | " | 2C | | " | " |
| | | | 22 | | | | | | | | " | | | | | | -12mA | | " | 2G | | " | " |
| | $I_{IL}$ | 3009 | 23 | | | | | | | | " | | | | | 0.4V | | | 5.5V | A | -0.7 | -1.6 | mA |
| | | | 24 | | | 0.4V | | | | | " | | | | | | | | " | B | " | " | " |
| | | | 25 | 0.4V | | | | | | | " | | | | | | | | " | 1C | " | " | " |
| | | | 26 | GND | 0.4V | | | | | | " | | | | | | | | " | 1G | " | " | " |
| | | | 27 | | | | | | | | " | | | | | | | 0.4V | " | 2C | " | " | " |
| | | | 28 | | | | | | | | " | | | | | | 0.4V | | " | 2G | " | " | " |
| | $I_{IH1}$ | 3010 | 29 | | | | | | | | " | | | | | 2.4V | | | " | A | | 60 | µA |
| | | | 30 | | | 2.4V | | | | | " | | | | | | | | " | B | | " | " |
| | | | 31 | 2.4V | | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 32 | 5.5V | 2.4V | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 33 | | | | | | | | " | | | | | | | 2.4V | " | 2C | | " | " |
| | | | 34 | | | | | | | | " | | | | | | 2.4V | | " | 2G | | " | " |
| | $I_{IH2}$ | | 35 | | | | | | | | " | | | | | 5.5V | | | " | A | | 150 | " |
| | | | 36 | | | 5.5V | | | | | " | | | | | | | | " | B | | " | " |
| | | | 37 | 5.5V | | | | | | | " | | | | | | | | " | 1C | | " | " |
| | | | 38 | 5.5V | 5.5V | | | | | | " | | | | | | | | " | 1G | | " | " |
| | | | 39 | | | | | | | | " | | | | | | | 5.5V | " | 2C | | " | " |
| | | | 40 | | | | | | | | " | | | | | | 5.5V | | " | 2G | | " | " |
| | $I_{CC}$ | 3005 | 41 | | GND | | | | | | " | | | | | | GND | GND | " | VCC | | 35 | mA |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |

See footnote at end of table.





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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 7 $T_C$=+25°C | Truth table test | 3014 | 42 | X | A | X | H | H | H | H | GND | H | H | H | H | X | A | X | 5.0V | 1/ | | | |
| | | | 43 | A | B | B | " | " | H | L | " | L | H | " | " | B | B | B | " | " | | | |
| | | | 44 | " | " | B | " | " | L | H | " | H | L | " | " | A | " | " | " | " | | | |
| | | | 45 | " | " | A | " | L | H | " | " | " | H | L | " | B | " | " | " | " | | | |
| | | | 46 | " | " | A | L | H | " | " | " | " | " | H | L | A | " | " | " | " | | | |
| | | | 47 | B | X | X | H | " | " | " | " | " | " | " | H | X | X | A | " | " | | | |
| | | | 48 | X | A | X | " | ' | " | " | " | " | " | " | " | X | A | X | " | " | | | |
| | | | 49 | B | B | B | " | " | " | " | " | L | " | " | " | B | B | B | " | " | | | |
| | | | 50 | " | " | B | " | " | " | " | " | H | L | " | " | A | " | " | " | " | | | |
| | | | 51 | " | " | A | " | " | " | " | " | " | H | L | " | B | " | " | " | " | | | |
| | | | 52 | " | " | A | " | " | " | " | " | " | " | H | L | A | " | " | " | " | | | |
| | | | 53 | A | " | B | " | " | " | L | " | " | " | " | H | B | " | A | " | " | | | |
| | | | 54 | " | " | B | " | " | L | H | " | " | " | " | " | A | " | " | " | " | | | |
| | | | 55 | " | " | A | " | L | H | " | " | " | " | " | " | B | " | " | " | " | | | |
| | | | 56 | " | " | A | L | H | H | " | " | " | " | " | " | A | " | " | " | " | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$=+25°C | $t_{PLH1}$ | 3003 (Fig 6) | 57 | 5.0V | IN | GND | | | | OUT | GND | | | | | GND | | | 5.0V | 1Y0 | 3 | 33 | ns |
| | | | 58 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 59 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 60 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 61 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 62 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 63 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 64 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PHL1}$ | | 65 | 5.0V | IN | GND | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 40 | " |
| | | | 66 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 67 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 68 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 69 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 70 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 71 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 72 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PLH2}$ | | 73 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 33 | " |
| | | | 74 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 75 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 76 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PHL2}$ | | 77 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 40 | " |
| | | | 78 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 79 | 5.0V | GND | 5.0V | | OUT | | | " | | | | | IN | | | " | 1Y2 | " | " | " |
| | | | 80 | | | 5.0V | | | | | " | | | OUT | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PLH3}$ | | 81 | | | GND | | | | | " | OUT | | | | GND | GND | IN | " | 2Y0 | " | 33 | " |
| | $t_{PHL3}$ | | 82 | | | " | | | | | " | OUT | | | | " | GND | IN | " | 2Y0 | " | 40 | " |
| | $t_{PLH4}$ | | 83 | IN | GND | " | | | | OUT | " | | | | | " | | | " | 1Y0 | " | 37 | " |
| | $t_{PHL4}$ | | 84 | IN | GND | " | | | | OUT | " | | | | | " | | | " | 1Y0 | " | 43 | " |

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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 9 $T_C$=+25°C | $t_{PLH5}$ | 3003 (Fig 6) | 85 | 5.0V | GND | IN | | OUT | | | GND | | | | | GND | | | 5.0V | 1Y2 | 3 | 44 | ns |
| | | | 86 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 87 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 88 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| | $t_{PHL5}$ | | 89 | 5.0V | GND | IN | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 90 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 91 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 92 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| 10 $T_C$=+125°C | $t_{PLH1}$ | | 93 | 5.0V | IN | " | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 43 | " |
| | | | 94 | " | " | | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 95 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 96 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 97 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 98 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 99 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 100 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PHL1}$ | | 101 | 5.0V | IN | GND | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 52 | " |
| | | | 102 | " | " | GND | | | OUT | | " | | | | | 5.0V | | | " | 1Y1 | " | " | " |
| | | | 103 | " | " | 5.0V | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 104 | " | " | 5.0V | OUT | | | | " | | | | | 5.0V | | | " | 1Y3 | " | " | " |
| | | | 105 | | | GND | | | | | " | OUT | | | | GND | IN | GND | " | 2Y0 | " | " | " |
| | | | 106 | | | GND | | | | | " | | OUT | | | 5.0V | " | " | " | 2Y1 | " | " | " |
| | | | 107 | | | 5.0V | | | | | " | | | OUT | | GND | " | " | " | 2Y2 | " | " | " |
| | | | 108 | | | 5.0V | | | | | " | | | | OUT | 5.0V | " | " | " | 2Y3 | " | " | " |
| | $t_{PLH2}$ | | 109 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 43 | " |
| | | | 110 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 111 | 5.0V | GND | 5.0V | | OUT | | | " | | | OUT | | IN | | | " | 1Y2 | " | " | " |
| | | | 112 | | | 5.0V | | | | | " | | | | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PHL2}$ | | 113 | 5.0V | GND | IN | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 52 | " |
| | | | 114 | | | IN | | | | | " | OUT | | | | GND | GND | GND | " | 2Y0 | " | " | " |
| | | | 115 | 5.0V | GND | 5.0V | | OUT | | | " | | | OUT | | IN | | | " | 1Y2 | " | " | " |
| | | | 116 | | | 5.0V | | | | | " | | | | | IN | GND | GND | " | 2Y2 | " | " | " |
| | $t_{PLH3}$ | | 117 | | | GND | | | | | " | OUT | | | | GND | GND | IN | " | 2Y0 | " | 43 | " |
| | $t_{PHL3}$ | | 118 | | | " | | | | | " | OUT | | | | GND | GND | IN | " | 2Y0 | " | 52 | " |
| | $t_{PLH4}$ | | 119 | IN | GND | " | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 48 | " |
| | $t_{PHL4}$ | | 120 | IN | GND | " | | | | OUT | " | | | | | GND | | | " | 1Y0 | " | 56 | " |

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TABLE III. Group A inspection for device type 03 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | 1C | 1G | B | 1Y3 | 1Y2 | 1Y1 | 1Y0 | GND | 2Y0 | 2Y1 | 2Y2 | 2Y3 | A | 2G | 2C | $V_{CC}$ | | Min | Max | |
| 10 $T_C$=+125°C | $t_{PLH5}$ | 3003 (Fig 6) | 121 | 5.0V | GND | IN | | OUT | | | GND | | | OUT | | GND | | | 5.0V | 1Y2 | 3 | 57 | ns |
| | | | 122 | | | IN | | | | | " | | | | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 123 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 124 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| | $t_{PHL5}$ | | 125 | 5.0V | GND | IN | | OUT | | | " | | | | | GND | | | " | 1Y2 | " | " | " |
| | | | 126 | | | IN | | | | | " | | | OUT | | GND | GND | GND | " | 2Y2 | " | " | " |
| | | | 127 | 5.0V | GND | GND | | | OUT | | " | | | | | IN | | | " | 1Y1 | " | " | " |
| | | | 128 | | | GND | | | | | " | | OUT | | | IN | GND | GND | " | 2Y1 | " | " | " |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

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TABLE III. Group A inspection for device type 04.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases A,B,D | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 1 | 2 | 3 | 4 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Case C | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | | Min | Max | |
| | | | Test no. | A | C | D | Y5 | Y6 | Y7 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | $V_{CC}$ | | | | |
| 1<br>$T_C$=+25°C | $V_{OH}$ | 3006 | 1 | 2.0V | 2.0V | 2.0V | | | | GND | | | | -.8mA | -.8mA | 2.0V | 4.5V | Y0 | 2.4 | | V |
| | | | 2 | " | " | " | | | | " | | | -.8mA | | | " | " | Y1 | " | | " |
| | | | 3 | " | " | " | | | | " | | -.8mA | | | | " | " | Y2 | " | | " |
| | | | 4 | " | " | " | | | | " | -.8mA | | | | | " | " | Y3 | " | | " |
| | | | 5 | " | " | " | | | | " | | | | | | " | " | Y4 | " | | " |
| | | | 6 | " | " | " | -.8mA | | | " | | | | | | " | " | Y5 | " | | " |
| | | | 7 | " | " | " | | -.8mA | | " | | | | | | " | " | Y6 | " | | " |
| | | | 8 | " | " | " | | | -.8mA | " | | | | | | " | " | Y7 | " | | " |
| | $V_{OL}$ | 3007 | 9 | 0.8V | 0.8V | 0.8V | | | | " | | | | | 16mA | 0.8V | " | Y0 | | 0.4 | " |
| | | | 10 | 2.0V | " | " | | | | " | | | 16mA | 16mA | | 0.8V | " | Y1 | | " | " |
| | | | 11 | 0.8V | " | " | | | | " | | 16mA | | | | 2.0V | " | Y2 | | " | " |
| | | | 12 | 2.0V | " | " | | | | " | 16mA | | | | | 2.0V | " | Y3 | | " | " |
| | | | 13 | 0.8V | 2.0V | " | | | | " | | | | | | 0.8V | " | Y4 | | " | " |
| | | | 14 | 2.0V | " | " | 16mA | | | " | | | | | | 0.8V | " | Y5 | | " | " |
| | | | 15 | 0.8V | " | " | | 16mA | | " | | | | | | 2.0V | " | Y6 | | " | " |
| | | | 16 | 2.0V | " | " | | | 16mA | " | | | | | | 2.0V | " | Y7 | | " | " |
| | $V_{IC}$ | | 17 | -12mA | | | | | | " | | | | | | | " | A | | -1.5 | " |
| | | | 18 | | -12mA | | | | | " | | | | | | -12mA | " | B | | " | " |
| | | | 19 | | | -12mA | | | | " | | | | | | | " | C | | " | " |
| | | | 20 | | | | | | | " | | | | | | | " | D | | " | " |
| | $I_{IL}$ | 3009 | 21 | 0.4V | | | | | | " | | | | | | | 5.5V | A | -.4 | -1.3 | mA |
| | | | 22 | | | | | | | " | | | | | | 0.4V | " | B | " | " | " |
| | | | 23 | | 0.4V | | | | | " | | | | | | | " | C | " | " | " |
| | | | 24 | | | 0.4V | | | | " | | | | | | | " | D | " | " | " |
| | $I_{IH1}$ | 3010 | 25 | 2.4V | | | | | | " | | | | | | | " | A | | 60 | µA |
| | | | 26 | | | | | | | " | | | | | | 2.4V | " | B | | " | " |
| | | | 27 | | 2.4V | | | | | " | | | | | | | " | C | | " | " |
| | | | 28 | | | 2.4V | | | | " | | | | | | | " | D | | " | " |
| | $I_{IH2}$ | | 29 | 5.5V | | | | | | " | | | | | | | " | A | | 150 | " |
| | | | 30 | | | | | | | " | | | | | | 5.5V | " | B | | " | " |
| | | | 31 | | 5.5V | | | | | " | | | | | | | " | C | | " | " |
| | | | 32 | | | 5.5V | | | | " | | | | | | | " | D | | " | " |
| | $I_{OS}$ | 3011 | 33 | 5.5V | 5.5V | " | | | | " | | | | GND | GND | 5.5V | " | Y0 | -10 | -55 | mA |
| | | | 34 | " | " | " | | | | " | | | GND | | | " | " | Y1 | " | " | " |
| | | | 35 | " | " | " | | | | " | | GND | | | | " | " | Y2 | " | " | " |
| | | | 36 | " | " | " | | | | " | GND | | | | | " | " | Y3 | " | " | " |
| | | | 37 | " | " | " | | | | " | | | | | | " | " | Y4 | " | " | " |
| | | | 38 | " | " | " | GND | | | " | | | | | | " | " | Y5 | " | " | " |
| | | | 39 | " | " | " | | GND | | " | | | | | | " | " | Y6 | " | " | " |
| | | | 40 | " | " | " | | | GND | " | | | | | | " | " | Y7 | " | " | " |
| | $I_{CC}$ | 3005 | 41 | | | GND | | | | " | | | | | | | " | | | 25 | " |

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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases A,B,D | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 1 | 2 | 3 | 4 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Case C | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | | Min | Max | |
| | | | Test no. | A | C | D | Y5 | Y6 | Y7 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | $V_{CC}$ | | | | |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$=+25°C | Truth table test | 3014 | 42 | B | B | B | H | H | H | GND | H | H | H | H | L | B | 5.0V | 1/ | | | |
| | | | 43 | A | " | " | " | " | " | " | " | " | H | L | H | B | " | " | | | |
| | | | 44 | B | " | " | " | " | " | " | " | " | L | H | " | A | " | " | | | |
| | | | 45 | A | " | " | " | " | " | " | " | L | H | " | " | A | " | " | | | |
| | | | 46 | B | A | " | " | " | " | " | L | H | " | " | " | B | " | " | | | |
| | | | 47 | A | " | " | L | " | " | " | H | " | " | " | " | B | " | " | | | |
| | | | 48 | B | " | " | H | L | " | " | " | " | " | " | " | A | " | " | | | |
| | | | 49 | A | " | " | " | H | L | " | " | " | " | " | " | A | " | " | | | |
| | | | 50 | B | B | A | " | " | H | " | " | " | " | " | " | B | " | " | | | |
| | | | 51 | A | " | " | " | " | " | " | " | " | " | " | " | B | " | " | | | |
| | | | 52 | B | " | " | " | " | " | " | " | " | " | " | " | A | " | " | | | |
| | | | 53 | A | " | " | " | " | " | " | " | " | " | " | " | A | " | " | | | |
| | | | 54 | B | A | " | " | " | " | " | " | " | " | " | " | B | " | " | | | |
| | | | 55 | A | " | " | " | " | " | " | " | " | " | " | " | B | " | " | | | |
| | | | 56 | B | " | " | " | " | " | " | " | " | " | " | " | A | " | " | | | |
| | | | 57 | A | " | " | " | " | " | " | " | " | " | " | " | A | " | " | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$=+25°C | $t_{PHL1}$ | 3003 (Fig 7) | 58 | IN | GND | GND | | | | GND | | | | OUT | OUT | GND | 5.0V | Y0 | 5 | 37 | ns |
| | | | 59 | 5.0V | GND | " | | | | " | | | | | | IN | " | Y1 | " | " | " |
| | | | 60 | GND | IN | " | | | | " | | | OUT | | | 5.0V | " | Y2 | " | " | " |
| | | | 61 | 5.0V | GND | IN | | | | " | | OUT | | | | 5.0V | " | Y3 | " | " | " |
| | | | 62 | IN | 5.0V | GND | | | | " | OUT | | | | | GND | " | Y4 | " | " | " |
| | | | 63 | 5.0V | " | " | OUT | | | " | | | | | | IN | " | Y5 | " | " | " |
| | | | 64 | IN | " | " | | OUT | | " | | | | | | 5.0V | " | Y6 | " | " | " |
| | | | 65 | 5.0V | " | " | | | OUT | " | | | | | | 5.0V | " | Y7 | " | " | " |
| | $t_{PLH1}$ | | 66 | IN | GND | GND | | | | " | | | | OUT | OUT | GND | " | Y0 | " | " | " |
| | | | 67 | 5.0V | GND | " | | | | " | | | | | | IN | " | Y1 | " | " | " |
| | | | 68 | GND | IN | " | | | | " | | | OUT | | | 5.0V | " | Y2 | " | " | " |
| | | | 69 | 5.0V | GND | IN | | | | " | | OUT | | | | 5.0V | " | Y3 | " | " | " |
| | | | 70 | IN | 5.0V | GND | | | | " | OUT | | | | | GND | " | Y4 | " | " | " |
| | | | 71 | 5.0V | " | " | OUT | | | " | | | | | | IN | " | Y5 | " | " | " |
| | | | 72 | IN | " | " | | OUT | | " | | | | | | 5.0V | " | Y6 | " | " | " |
| | | | 73 | 5.0V | " | " | | | OUT | " | | | | | | 5.0V | " | Y7 | " | " | " |
| | $t_{PHL2}$ | | 74 | IN | GND | GND | | | | " | | | | OUT | | GND | " | Y1 | " | 42 | " |
| | | | 75 | 5.0V | GND | " | | | | " | | OUT | | | | IN | " | Y3 | " | " | " |
| | | | 76 | 5.0V | IN | " | OUT | | | " | | | | | | GND | " | Y5 | " | " | " |
| | $t_{PLH2}$ | | 77 | IN | GND | GND | | | | " | | | | OUT | | GND | " | Y1 | " | " | " |
| | | | 78 | 5.0V | GND | " | | | | " | | OUT | | | | IN | " | Y3 | " | " | " |
| | | | 79 | 5.0V | IN | " | OUT | | | " | | | | | | GND | " | Y5 | " | " | " |

See footnote at end of table.





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TABLE III. Group A inspection for device type 04 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases A,B,D | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 1 | 2 | 3 | 4 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Case C | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | | Min | Max | |
| | | | Test no. | A | C | D | Y5 | Y6 | Y7 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | $V_{CC}$ | | | | |
| 10 $T_C$=+125°C | $t_{PHL1}$ | 3003 (Fig 7) | 80 | IN | GND | GND | | | | GND | | | | | OUT | GND | 5.0V | Y0 | 5 | 48 | ns |
| | | | 81 | 5.0V | GND | " | | | | " | | | | OUT | | IN | " | Y1 | " | " | " |
| | | | 82 | GND | IN | " | | | | " | | | OUT | | | 5.0V | " | Y2 | " | " | " |
| | | | 83 | 5.0V | GND | IN | | | | " | | OUT | | | | 5.0V | " | Y3 | " | " | " |
| | | | 84 | IN | 5.0V | GND | | | | " | OUT | | | | | GND | " | Y4 | " | " | " |
| | | | 85 | 5.0V | " | " | OUT | | | " | | | | | | IN | " | Y5 | " | " | " |
| | | | 86 | IN | " | " | | OUT | | " | | | | | | 5.0V | " | Y6 | " | " | " |
| | | | 87 | 5.0V | " | IN | | | OUT | " | | | | | | 5.0V | " | Y7 | " | " | " |
| | $t_{PLH1}$ | | 88 | IN | GND | GND | | | | " | | | | | OUT | GND | " | Y0 | " | " | " |
| | | | 89 | 5.0V | GND | " | | | | " | | | | OUT | | IN | " | Y1 | " | " | " |
| | | | 90 | GND | IN | " | | | | " | | | OUT | | | 5.0V | " | Y2 | " | " | " |
| | | | 91 | 5.0V | GND | IN | | | | " | | OUT | | | | 5.0V | " | Y3 | " | " | " |
| | | | 92 | IN | 5.0V | GND | | | | " | OUT | | | | | GND | " | Y4 | " | " | " |
| | | | 93 | 5.0V | " | " | OUT | | | " | | | | | | IN | " | Y5 | " | " | " |
| | | | 94 | IN | " | " | | OUT | | " | | | | | | 5.0V | " | Y6 | " | " | " |
| | | | 95 | 5.0V | " | IN | | | OUT | " | | | | | | 5.0V | " | Y7 | " | " | " |
| | $t_{PHL2}$ | | 96 | IN | GND | GND | | | | " | | | | OUT | | GND | " | Y1 | " | 53 | " |
| | | | 97 | 5.0V | GND | " | | | | " | | OUT | | | | IN | " | Y3 | " | " | " |
| | | | 98 | 5.0V | IN | " | OUT | | | " | | | | | | GND | " | Y5 | " | " | " |
| | $t_{PLH2}$ | | 99 | IN | GND | " | | | | " | | | | OUT | | GND | " | Y1 | " | " | " |
| | | | 100 | 5.0V | GND | " | | | | " | | OUT | | | | IN | " | Y3 | " | " | " |
| | | | 101 | 5.0V | IN | " | OUT | | | " | | | | | | GND | " | Y5 | " | " | " |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | |

See footnote at end of table.





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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 1<br>$T_C$=+25°C | $V_{OH}$ | 3006 | 1 | 2.0V | 2.0V | | | | | | GND | | | | -.8mA | | 2.0V | 2.0V | 4.5V | Y0 | 2.4 | | V |
| | | | 2 | " | " | | | | | | " | | | | | | " | " | " | Y1 | " | | " |
| | | | 3 | " | " | | | | | | " | | | -.8mA | | | " | " | " | Y2 | " | | " |
| | | | 4 | " | " | | | | | | " | | -.8mA | | | | " | " | " | Y3 | " | | " |
| | | | 5 | " | " | | | | | | " | -.8mA | | | | | " | " | " | Y4 | " | | " |
| | | | 6 | " | " | -.8mA | | | | | " | | | | | | " | " | " | Y5 | " | | " |
| | | | 7 | " | " | | -.8mA | | | | " | | | | | | " | " | " | Y6 | " | | " |
| | | | 8 | " | " | | | -.8mA | | | " | | | | | | " | " | " | Y7 | " | | " |
| | | | 9 | " | " | | | | -.8mA | | " | | | | | | " | " | " | Y8 | " | | " |
| | | | 10 | " | " | | | | | -.8mA | " | | | | | | " | 0.8V | " | Y9 | " | | " |
| | $V_{OL}$ | 3007 | 11 | 0.8V | 0.8V | | | | | | " | | | | | | 0.8V | 0.8V | " | Y0 | | 0.4 | " |
| | | | 12 | " | " | | | | | | " | | | | 16mA | | 0.8V | 2.0V | " | Y1 | | " | " |
| | | | 13 | " | " | | | | | | " | | | 16mA | | | 2.0V | 0.8V | " | Y2 | | " | " |
| | | | 14 | " | " | | | | | | " | | 16mA | | | | 2.0V | 2.0V | " | Y3 | | " | " |
| | | | 15 | 2.0V | " | | | | | | " | 16mA | | | | | 0.8V | 0.8V | " | Y4 | | " | " |
| | | | 16 | " | " | 16mA | | | | | " | | | | | | 0.8V | 2.0V | " | Y5 | | " | " |
| | | | 17 | " | " | | 16mA | | | | " | | | | | | 2.0V | 0.8V | " | Y6 | | " | " |
| | | | 18 | " | " | | | 16mA | | | " | | | | | | 2.0V | 2.0V | " | Y7 | | " | " |
| | | | 19 | 0.8V | 2.0V | | | | 16mA | | " | | | | | | 0.8V | 0.8V | " | Y8 | | " | " |
| | | | 20 | 0.8V | 2.0V | | | | | 16mA | " | | | | | | 0.8V | 2.0V | " | Y9 | | " | " |
| | $V_{IC}$ | | 21 | | | | | | | | " | | | | | | | -12mA | " | A | | -1.5 | " |
| | | | 22 | | | | | | | | " | | | | | | -12mA | " | " | B | | " | " |
| | | | 23 | -12mA | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 24 | | -12mA | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{IL}$ | 3009 | 25 | | | | | | | | " | | | | | | | 0.4V | 5.5V | A | -0.4 | -1.3 | mA |
| | | | 26 | | | | | | | | " | | | | | | 0.4V | " | " | B | " | " | " |
| | | | 27 | 0.4V | | | | | | | " | | | | | | | " | " | C | " | " | " |
| | | | 28 | | 0.4V | | | | | | " | | | | | | | " | " | D | " | " | " |
| | $I_{IH1}$ | 3010 | 29 | | | | | | | | " | | | | | | | 2.4V | " | A | | 60 | µA |
| | | | 30 | | | | | | | | " | | | | | | 2.4V | " | " | B | | " | " |
| | | | 31 | 2.4V | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 32 | | 2.4V | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{IH2}$ | | 33 | | | | | | | | " | | | | | | | 5.5V | " | A | | 150 | " |
| | | | 34 | | | | | | | | " | | | | | | 5.5V | " | " | B | | " | " |
| | | | 35 | 5.5V | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 36 | | 5.5V | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{OS}$ | 3011 | 37 | 5.5V | " | | | | | | " | | | | GND | GND | 5.5V | 5.5V | " | Y0 | -10 | -55 | mA |
| | | | 38 | " | " | | | | | | " | | | GND | | | " | " | " | Y1 | " | " | " |
| | | | 39 | " | " | | | | | | " | | GND | | | | " | " | " | Y2 | " | " | " |
| | | | 40 | " | " | | | | | | " | GND | | | | | " | " | " | Y3 | " | " | " |
| | | | 41 | " | " | GND | | | | | " | | | | | | " | " | " | Y4 | " | " | " |
| | | | 42 | " | " | | GND | | | | " | | | | | | " | " | " | Y5 | " | " | " |
| | | | 43 | " | " | | | GND | | | " | | | | | | " | " | " | Y6 | " | " | " |
| | | | 44 | " | " | | | | GND | | " | | | | | | " | " | " | Y7 | " | " | " |
| | | | 45 | " | " | | | | | GND | " | | | | | | " | " | " | Y8 | " | " | " |
| | | | 46 | " | " | | | | | | " | | | | | | " | GND | " | Y9 | " | " | " |
| | $I_{CC}$ | 3005 | 47 | | GND | | | | | | " | | | | | | | | " | VCC | | 34 | " |

See footnote at end of table.



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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$=+25°C | Truth table test | 3014 | 48 | B | B | H | H | H | H | H | GND | H | H | H | H | L | B | B | 5.0V | 1/ | | | |
| | | | 49 | " | " | " | " | " | " | " | " | " | " | H | L | H | B | A | " | " | | | |
| | | | 50 | " | " | " | " | " | " | " | " | " | " | L | H | " | A | B | " | " | | | |
| | | | 51 | " | " | " | " | " | " | " | " | " | L | H | " | " | A | A | " | " | | | |
| | | | 52 | A | " | " | " | " | " | " | " | L | H | " | " | " | B | B | " | " | | | |
| | | | 53 | " | " | L | " | " | " | " | " | H | " | " | " | " | B | A | " | " | | | |
| | | | 54 | " | " | H | L | " | " | " | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 55 | " | " | " | H | L | " | " | " | " | " | " | " | " | A | A | " | " | | | |
| | | | 56 | B | A | " | " | H | L | " | " | " | " | " | " | " | B | B | " | " | | | |
| | | | 57 | " | " | " | " | " | H | L | " | " | " | " | " | " | B | A | " | " | | | |
| | | | 58 | " | " | " | " | " | L | H | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 59 | " | " | " | " | " | H | L | " | " | " | " | " | " | A | A | " | " | | | |
| | | | 60 | A | " | " | " | " | L | H | " | " | " | " | " | " | B | B | " | " | | | |
| | | | 61 | " | " | " | " | " | H | L | " | " | " | " | " | " | B | A | " | " | | | |
| | | | 62 | " | " | " | " | " | L | H | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 63 | " | " | " | " | " | H | L | " | " | " | " | " | " | A | A | " | " | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$=+25°C | $t_{PHL1}$ | 3003 (Fig 7) | 64 | GND | GND | OUT | OUT | OUT | OUT | OUT | GND | OUT | OUT | OUT | OUT | OUT | GND | IN | 5.0V | Y0 | 5 | 37 | ns |
| | | | 65 | GND | " | | | | | | " | | | | | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 66 | IN | " | | | | | | " | | | OUT | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 67 | GND | IN | | | | | | " | | OUT | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 68 | 5.0V | GND | | | | | | " | OUT | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 69 | " | GND | | | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 70 | " | IN | | | | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 71 | " | IN | | | | | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 72 | GND | 5.0V | | | | | | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 73 | " | 5.0V | | | | | | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PLH1}$ | | 74 | " | GND | | | | | | " | | | | | | GND | IN | " | Y0 | " | " | " |
| | | | 75 | " | " | | | | | | " | | | | | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 76 | IN | " | | | | | | " | | | | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 77 | GND | IN | | | | | | " | | | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 78 | 5.0V | GND | | | | | | " | | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 79 | " | GND | | | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 80 | " | IN | | | | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 81 | " | IN | | | | | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 82 | GND | 5.0V | | | | | | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 83 | " | 5.0V | | | | | | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PLH2}$ | | 84 | " | GND | | | | | | " | | | | | | GND | IN | " | Y1 | " | 42 | " |
| | | | 85 | " | " | | | | | | " | | | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 86 | IN | " | | | | | | " | | | | | | GND | " | " | Y5 | " | " | " |
| | | | 87 | 5.0V | IN | | | | | | " | | | | | | 5.0V | " | " | Y9 | " | " | " |
| | $t_{PLH2}$ | | 88 | GND | GND | | | | | | " | | | | | | GND | IN | " | Y1 | " | " | " |
| | | | 89 | GND | " | | | | | | " | | | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 90 | IN | " | | | | | | " | | | | | | GND | " | " | Y5 | " | " | " |
| | | | 91 | 5.0V | IN | | | | | | " | | | | | | 5.0V | " | " | Y9 | " | " | " |

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TABLE III. Group A inspection for device type 05 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 10<br>$T_C$=+125°C | $t_{PHL1}$ | 3003<br>(Fig 7) | 92<br>93<br>94<br>95<br>96<br>97<br>98<br>99<br>100<br>101 | GND<br>GND<br>IN<br>GND<br>5.0V<br>“<br>“<br>“<br>GND<br>“ | GND<br>“<br>“<br>IN<br>GND<br>GND<br>IN<br>IN<br>5.0V<br>5.0V | OUT | OUT | OUT | OUT | OUT | GND<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | OUT | OUT | OUT | OUT | OUT | GND<br>IN<br>5.0V<br>5.0V<br>GND<br>IN<br>5.0V<br>5.0V<br>GND<br>IN | IN<br>5.0V<br>GND<br>5.0V<br>IN<br>5.0V<br>GND<br>5.0V<br>IN<br>5.0V | 5.0V<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | Y0<br>Y1<br>Y2<br>Y3<br>Y4<br>Y5<br>Y6<br>Y7<br>Y8<br>Y9 | 5<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | 48<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | ns<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ |
| | $t_{PLH1}$ | | 102<br>103<br>104<br>105<br>106<br>107<br>108<br>109<br>110<br>111 | “<br>“<br>IN<br>GND<br>5.0V<br>“<br>“<br>“<br>GND<br>“ | GND<br>“<br>“<br>IN<br>GND<br>GND<br>IN<br>IN<br>5.0V<br>5.0V | OUT | OUT | OUT | OUT | OUT | “<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | OUT | OUT | OUT | OUT | OUT | GND<br>IN<br>5.0V<br>5.0V<br>GND<br>IN<br>5.0V<br>5.0V<br>GND<br>IN | IN<br>5.0V<br>GND<br>5.0V<br>IN<br>5.0V<br>GND<br>5.0V<br>IN<br>5.0V | “<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | Y0<br>Y1<br>Y2<br>Y3<br>Y4<br>Y5<br>Y6<br>Y7<br>Y8<br>Y9 | “<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | “<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ | “<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“<br>“ |
| | $t_{PHL2}$ | | 112<br>113<br>114<br>115 | “<br>“<br>IN<br>5.0V | GND<br>“<br>“<br>IN | OUT | | | | OUT | “<br>“<br>“<br>“ | | OUT | | OUT | | GND<br>IN<br>GND<br>5.0V | IN<br>5.0V<br>“<br>“ | “<br>“<br>“<br>“ | Y1<br>Y3<br>Y5<br>Y9 | “<br>“<br>“<br>“ | 53<br>“<br>“<br>“ | “<br>“<br>“<br>“ |
| | $t_{PLH2}$ | | 116<br>117<br>118<br>119 | GND<br>GND<br>IN<br>5.0V | GND<br>“<br>“<br>IN | OUT | | | | OUT | “<br>“<br>“<br>“ | | OUT | | OUT | | GND<br>IN<br>GND<br>5.0V | IN<br>5.0V<br>“<br>“ | “<br>“<br>“<br>“ | Y1<br>Y3<br>Y5<br>Y9 | “<br>“<br>“<br>“ | “<br>“<br>“<br>“ | “<br>“<br>“<br>“ |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

See footnote at end of table.





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TABLE III. Group A inspection for device type 06.

Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 1 $T_C$=+25°C | $V_{OH}$ | 3006 | 1 | 2.0V | 2.0V | | | | | | GND | | | | -.8mA | | 2.0V | 2.0V | 4.5V | Y0 | 2.4 | | V |
| | | | 2 | " | " | | | | | | " | | | | " | | " | " | " | Y1 | " | | " |
| | | | 3 | " | " | | | | | | " | | | -.8mA | " | | " | " | " | Y2 | " | | " |
| | | | 4 | " | " | | | | | | " | | -.8mA | " | " | | " | " | " | Y3 | " | | " |
| | | | 5 | " | " | | | | | | " | -.8mA | " | " | " | | " | " | " | Y4 | " | | " |
| | | | 6 | " | " | -.8mA | | | | | " | " | " | " | " | | " | " | " | Y5 | " | | " |
| | | | 7 | " | " | | -.8mA | | | | " | " | " | " | " | | " | " | " | Y6 | " | | " |
| | | | 8 | " | " | | | -.8mA | | | " | " | " | " | " | | " | " | " | Y7 | " | | " |
| | | | 9 | " | " | | | | -.8mA | | " | " | " | " | " | | " | " | " | Y8 | " | | " |
| | | | 10 | " | " | | | | | -.8mA | " | " | " | " | " | | " | " | " | Y9 | " | | " |
| | $V_{OL}$ | 3007 | 11 | 0.8V | 0.8V | | | | | | " | | | | | | 0.8V | 0.8V | " | Y0 | | 0.4 | " |
| | | | 12 | " | " | | | | | | " | | | | 16mA | | 0.8V | 2.0V | " | Y1 | | " | " |
| | | | 13 | " | " | | | | | | " | | | 16mA | " | | 2.0V | 0.8V | " | Y2 | | " | " |
| | | | 14 | " | " | | | | | | " | | 16mA | " | " | | 2.0V | 2.0V | " | Y3 | | " | " |
| | | | 15 | 2.0V | " | | | | | | " | 16mA | " | " | " | | 0.8V | 0.8V | " | Y4 | | " | " |
| | | | 16 | " | " | 16mA | | | | | " | " | " | " | " | | 0.8V | 2.0V | " | Y5 | | " | " |
| | | | 17 | " | " | | 16mA | | | | " | " | " | " | " | | 2.0V | 0.8V | " | Y6 | | " | " |
| | | | 18 | " | " | | | 16mA | | | " | " | " | " | " | | 2.0V | 2.0V | " | Y7 | | " | " |
| | | | 19 | 0.8V | 2.0V | | | | 16mA | | " | " | " | " | " | | 0.8V | 0.8V | " | Y8 | | " | " |
| | | | 20 | 0.8V | 2.0V | | | | | 16mA | " | " | " | " | " | | 0.8V | 2.0V | " | Y9 | | " | " |
| | $V_{IC}$ | | 21 | | | | | | | | " | | | | | | | -12mA | " | A | | -1.5 | " |
| | | | 22 | | | | | | | | " | | | | | | -12mA | " | " | B | | " | " |
| | | | 23 | -12mA | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 24 | | -12mA | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{IL}$ | 3009 | 25 | | | | | | | | " | | | | | | | 0.4V | 5.5V | A | -0.7 | -1.6 | mA |
| | | | 26 | | | | | | | | " | | | | | | 0.4V | " | " | B | " | " | " |
| | | | 27 | 0.4V | | | | | | | " | | | | | | | " | " | C | " | " | " |
| | | | 28 | | 0.4V | | | | | | " | | | | | | | " | " | D | " | " | " |
| | $I_{IH1}$ | 3010 | 29 | | | | | | | | " | | | | | | | 2.4V | " | A | | 60 | µA |
| | | | 30 | | | | | | | | " | | | | | | 2.4V | " | " | B | | " | " |
| | | | 31 | 2.4V | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 32 | | 2.4V | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{IH2}$ | | 33 | | | | | | | | " | | | | | | | 5.5V | " | A | | 150 | " |
| | | | 34 | | | | | | | | " | | | | | | 5.5V | " | " | B | | " | " |
| | | | 35 | 5.5V | | | | | | | " | | | | | | | " | " | C | | " | " |
| | | | 36 | | 5.5V | | | | | | " | | | | | | | " | " | D | | " | " |
| | $I_{OS}$ | 3011 | 37 | 5.5V | " | | | | | | " | | | | | GND | 5.5V | 5.5V | " | Y0 | -10 | -55 | mA |
| | | | 38 | " | " | | | | | | " | | | | GND | | " | " | " | Y1 | " | " | " |
| | | | 39 | " | " | | | | | | " | | | GND | " | | " | " | " | Y2 | " | " | " |
| | | | 40 | " | " | | | | | | " | | GND | " | " | | " | " | " | Y3 | " | " | " |
| | | | 41 | " | " | | | | | | " | GND | " | " | " | | " | " | " | Y4 | " | " | " |
| | | | 42 | " | " | GND | | | | | " | " | " | " | " | | " | " | " | Y5 | " | " | " |
| | | | 43 | " | " | | GND | | | | " | " | " | " | " | | " | " | " | Y6 | " | " | " |
| | | | 44 | " | " | | | GND | | | " | " | " | " | " | | " | " | " | Y7 | " | " | " |
| | | | 45 | " | " | | | | GND | | " | " | " | " | " | | " | " | " | Y8 | " | " | " |
| | | | 46 | " | " | | | | | GND | " | " | " | " | " | | " | " | " | Y9 | " | " | " |
| | $I_{CC}$ | 3005 | 47 | | GND | | | | | | " | | | | | | | | " | VCC | | 44 | " |

See footnote at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 2 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = +125°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Same tests, terminal conditions, and limits as subgroup 1, except $T_C$ = -55°C and $V_{IC}$ tests are omitted. | | | | | | | | | | | | | | | | | | | | | | |
| 7 $T_C$=+25°C | Truth table test | 3014 | 48 | B | B | H | H | H | H | H | GND | H | H | H | H | L | B | B | 5.0V | 1/ | | | |
| | | | 49 | " | " | " | " | " | " | " | " | " | " | H | L | H | B | A | " | " | | | |
| | | | 50 | " | " | " | " | " | " | " | " | " | L | L | H | " | A | B | " | " | | | |
| | | | 51 | " | " | " | " | " | " | " | " | " | " | H | " | " | A | A | " | " | | | |
| | | | 52 | A | " | " | " | " | " | " | " | L | L | H | " | " | B | B | " | " | | | |
| | | | 53 | " | " | L | " | " | " | " | " | H | " | " | " | " | B | A | " | " | | | |
| | | | 54 | " | " | H | L | " | " | " | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 55 | " | " | " | H | L | " | " | " | " | " | " | " | " | A | A | " | " | | | |
| | | | 56 | B | A | " | " | H | L | " | " | " | " | " | " | " | B | B | " | " | | | |
| | | | 57 | " | " | " | " | " | H | L | " | " | " | " | " | " | B | A | " | " | | | |
| | | | 58 | " | " | " | " | " | " | H | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 59 | " | " | " | " | " | " | " | " | " | " | " | " | " | A | A | " | " | | | |
| | | | 60 | A | " | " | " | " | " | " | " | " | " | " | " | " | B | B | " | " | | | |
| | | | 61 | " | " | " | " | " | " | " | " | " | " | " | " | " | B | A | " | " | | | |
| | | | 62 | " | " | " | " | " | " | " | " | " | " | " | " | " | A | B | " | " | | | |
| | | | 63 | " | " | " | " | " | " | " | " | " | " | " | " | " | A | A | " | " | | | |
| 8 | Repeat subgroup 7 at $T_C$ = +125°C and $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |
| 9 $T_C$=+25°C | $t_{PHL1}$ | 3003 (Fig 7) | 64 | GND | GND | | | | | | GND | | | | | OUT | GND | IN | 5.0V | Y0 | 5 | 37 | ns |
| | | | 65 | GND | " | | | | | | " | | | | OUT | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 66 | IN | " | | | | | | " | | | OUT | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 67 | GND | IN | | | | | | " | | OUT | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 68 | 5.0V | GND | | | | | | " | OUT | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 69 | " | GND | OUT | | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 70 | " | IN | | OUT | | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 71 | " | IN | | | OUT | | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 72 | GND | 5.0V | | | | OUT | | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 73 | " | 5.0V | | | | | OUT | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PLH1}$ | | 74 | " | GND | | | | | | " | | | | OUT | OUT | GND | IN | " | Y0 | " | " | " |
| | | | 75 | " | " | | | | | | " | | | OUT | | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 76 | IN | " | | | | | | " | | OUT | | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 77 | GND | IN | | | | | | " | OUT | | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 78 | 5.0V | GND | OUT | | | | | " | | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 79 | " | GND | | OUT | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 80 | " | IN | | | OUT | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 81 | " | IN | | | | OUT | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 82 | GND | 5.0V | | | | | OUT | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 83 | " | 5.0V | | | | | | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PHL2}$ | | 84 | " | GND | | | | | | " | | | | OUT | | GND | IN | " | Y1 | " | 42 | " |
| | | | 85 | " | " | | | | | | " | | OUT | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 86 | IN | " | OUT | | | | | " | | | | | | GND | 5.0V | " | Y5 | " | " | " |
| | | | 87 | GND | IN | | | | | OUT | " | | | | | | 5.0V | GND | " | Y9 | " | " | " |
| | $t_{PLH2}$ | | 88 | " | GND | | | | | | " | | | | OUT | | GND | IN | " | Y1 | " | " | " |
| | | | 89 | " | " | | | | | | " | | OUT | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 90 | IN | " | OUT | | | | | " | | | | | | GND | 5.0V | " | Y5 | " | " | " |
| | | | 91 | GND | IN | | | | | OUT | " | | | | | | 5.0V | GND | " | Y9 | " | " | " |

See footnote at end of table.





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TABLE III. Group A inspection for device type 06 – Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

| Subgroup | Symbol | MIL-STD-883 method | Cases E,F | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | Measured terminal | Limits | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Test no. | C | D | Y5 | Y6 | Y7 | Y8 | Y9 | GND | Y4 | Y3 | Y2 | Y1 | Y0 | B | A | $V_{CC}$ | | Min | Max | |
| 10 $T_C$=+125°C | $t_{PHL1}$ | 3003 (Fig 7) | 92 | GND | GND | | | | | | GND | | | | OUT | OUT | GND | IN | 5.0V | Y0 | 5 | 48 | ns |
| | | | 93 | GND | " | | | | | | " | | | OUT | | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 94 | IN | " | | | | | | " | | OUT | | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 95 | GND | IN | | | | | | " | | | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 96 | 5.0V | GND | | | | | | " | OUT | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 97 | " | GND | OUT | | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 98 | " | IN | | OUT | | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 99 | " | IN | | | OUT | | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 100 | GND | 5.0V | | | | OUT | | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 101 | " | 5.0V | | | | | OUT | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PLH1}$ | | 102 | " | GND | | | | | | " | | | | | OUT | GND | IN | " | Y0 | " | " | " |
| | | | 103 | " | " | | | | | | " | | | | OUT | | IN | 5.0V | " | Y1 | " | " | " |
| | | | 104 | IN | " | | | | | | " | | | OUT | | | 5.0V | GND | " | Y2 | " | " | " |
| | | | 105 | GND | IN | | | | | | " | | OUT | | | | 5.0V | 5.0V | " | Y3 | " | " | " |
| | | | 106 | 5.0V | GND | | | | | | " | OUT | | | | | GND | IN | " | Y4 | " | " | " |
| | | | 107 | " | GND | OUT | | | | | " | | | | | | IN | 5.0V | " | Y5 | " | " | " |
| | | | 108 | " | IN | | OUT | | | | " | | | | | | 5.0V | GND | " | Y6 | " | " | " |
| | | | 109 | " | IN | | | OUT | | | " | | | | | | 5.0V | 5.0V | " | Y7 | " | " | " |
| | | | 110 | GND | 5.0V | | | | OUT | | " | | | | | | GND | IN | " | Y8 | " | " | " |
| | | | 111 | " | 5.0V | | | | | OUT | " | | | | | | IN | 5.0V | " | Y9 | " | " | " |
| | $t_{PHL2}$ | | 112 | " | GND | | | | | | " | | | | OUT | | GND | IN | " | Y1 | " | 53 | " |
| | | | 113 | " | " | | | | | | " | | OUT | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 114 | IN | " | OUT | | | | | " | | | | | | GND | " | " | Y5 | " | " | " |
| | | | 115 | GND | IN | | | | | OUT | " | | | | | | " | " | " | Y9 | " | " | " |
| | $t_{PLH2}$ | | 116 | " | GND | | | | | | " | | | | OUT | | " | IN | " | Y1 | " | " | " |
| | | | 117 | " | " | | | | | | " | | OUT | | | | IN | 5.0V | " | Y3 | " | " | " |
| | | | 118 | IN | " | OUT | | | | | " | | | | | | GND | " | " | Y5 | " | " | " |
| | | | 119 | GND | IN | | | | | OUT | " | | | | | | GND | " | " | Y9 | " | " | " |
| 11 | Same tests, terminal conditions, and limits as subgroup 10, except $T_C$ = -55°C. | | | | | | | | | | | | | | | | | | | | | | |

1/ Inputs: A = 2 V minimum and B = 0.8 V maximum.
Input: X = A or B
Outputs: Output voltages shall be either:
(a) H = 2.4 V and L = 0.4 V when using a high speed checker double comparator.
(b) H ≥ 1.5 V and L < 1.5 V when using a high speed checker single comparator.




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4. VERIFICATION.

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as function as described herein.

4.2 Screening. Screening shall be in accordance with MIL-PRF-38535, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. The burn-in test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1015 of MIL-STD-883.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. Additional screening for space level product shall be as specified in MIL-PRF-38535.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-38535.

4.4 Technology Conformance inspection (TCI). Technology conformance inspection shall be in accordance with MIL-PRF-38535 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table III of MIL-PRF-38535 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of MIL-PRF-38535.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table IV of MIL-PRF-38535 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. The steady-state life test duration, test condition, and test temperature, or approved alternatives shall be as specified in the device manufacturer's QM plan in accordance with MIL-PRF-38535. The burn-in test circuit shall be maintained under document control by the device manufacturer's Technology Review Board (TRB) in accordance with MIL-PRF-38535 and shall be made available to the acquiring or preparing activity upon request. The test circuit shall specify the inputs, outputs, biases, and power dissipation, as applicable, in accordance with the intent specified in test method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table V of MIL-PRF-38535. End point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified and as follows.

4.5.1 Voltage and current. All voltage values given are referenced to the microcircuit ground terminals. Currents given are conventional current and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.



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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Complete part number (see 1.2).

c. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

d. Requirements for certificate of compliance, if applicable.

e. Requirements for notification of change of product or process to acquiring activity in addition to notification of the qualifying activity, if applicable.

f. Requirements for failure analysis (including required test condition of MIL-STD-883, method 5003), corrective action and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements should not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

i. Requirements for "JAN" marking.

j. Packaging requirements (see 5.1).

6.3 Superseding information. The requirements of MIL-M-38510 have been superseded to take advantage of the available Qualified Manufacturer Listing (QML) system provided by MIL-PRF-38535. Previous references to MIL-M-38510 in this document have been replaced by appropriate references to MIL-PRF-38535. All technical requirements now consist of this specification and MIL-PRF-38535. The MIL-M-38510 specification sheet number and PIN have been retained to avoid adversely impacting existing government logistics systems and contractor's parts lists.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-38535 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DSCC-VQ, 3990 E. Broad Street, Columbus, Ohio 43128-3990.

6.5 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-PRF-38535 and MIL-STD-1331, and as follows:.

| | | |
|---|---|---|
| GND | ..... | Ground zero voltage potential. |
| $V_{IN}$ | ..... | Voltage level at an input terminal. |
| $V_{IC}$ | ..... | Input clamp voltage. |
| $I_{IN}$ | ..... | Current flowing into an input terminal. |



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6.6 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish A (see 3.4). Longer length leads and lead forming should not affect the part number.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information should not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-PRF-38535.

| Military device type | Generic-industry type |
|---|---|
| 01 | 54154, 9311 |
| 02 | 54155 |
| 03 | 54156 |
| 04 | 8250 |
| 05 | 8251 |
| 06 | 8252, 9301 |

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army – CR
Navy - EC
Air Force - 11
NASA - NA
DLA – CC

Review activities:
Army - MI, SM
Navy - AS, CG, MC, SH, TD
Air Force – 03, 19, 99

Preparing activity:
DLA - CC

Project 5962-2097

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil .



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